

A Multimodal Machine Learning Framework for Sustainable Crop Decision Support in Bangladesh: Integrating Soil Imagery, Geospatial Sensing, and Climatic Data

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A thesis submitted to the Department of Computer Science and Engineering
in partial fulfillment of the requirements for the degree of
B.Sc. in Computer Science and Engineering

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Declaration

It is hereby declared that

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3. The thesis does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
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Abstract

Bangladesh’s agriculture land management is facing unprecedented climate volatility and soil degradation, which are forcing Bangladesh to take multi-source data into account for proper decision making on a local basis. A cascaded multimodal machine learning approach has been proposed to classify soil conditions and predict Aman rice and Wheat yields at several districts. This study starts with the fusion of satellite, climate and historical yield data, and a preparation of a soil image manifest in balance. Two parallel machine learning pipelines are then trained. The raw imagery is classified into local soil types using a strong macro F1 score of 0.870 Convolutional Neural Network (EfficientNet-B0). XGBoost ensemble models are built for each crop to predict the yield of each crop. The yield models have a strong potential of prediction on a temporal holdout with an R^2 of 0.847 for Aman rice and 0.512 for Wheat. The key challenge is the scarcity of paired image-yield datasets. Thus, the system uses decision-level fusion (DLF) to readily integrate the CNN soil predictions with the XGBoost yield predictions, guided by agronomic rules. Last, this multimodal application is implemented using a Streamlit interactive GUI for actionable support to stakeholders’ decision making.

Keywords: Multimodal machine learning, soil image classification, crop yield forecasting, decision-level fusion, Bangladesh agriculture, EfficientNet-B0, XGBoost, remote sensing

Dedication (Optional)

A dedication is the expression of friendly connection or thanks by the author towards another person. It can occupy one or multiple lines depending on its importance. You can remove this page if you want.

Acknowledgement

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Chapter 1

Introduction

Although agriculture is the socio-economic foundation of Bangladesh, its production system is very sensitive to seasonal changes in climatic conditions, unpredictable weather situations and diverse soil types in different regions of the country. Small-holder farmers have always used their practical knowledge that spans several generations for crop planning, but climate change has added to the uncertainty of farming, especially in terms of the planned production of important crops such as aman rice and wheat. The digital revolution has also created a vast amount of agricultural data, including time-series of historical climate information, gridded soil properties, and satellite-based remote sensing data, but these data are mostly fragmented and applied in isolation. This thesis tackles this fragmentation by proposing a modular, decision level multimodal machine learning framework tailored for the agriculture context of Bangladesh. This research develops a knowledge-driven, rule-based, interpretable, and scalable recommendation system that connects ground-level computer vision soil classification with macro-level geospatial and climatic tabular yield prediction, thereby shifting the state of agriculture from a reactive one of uncertainty to a sustainable one of scalable and interpretable resilience.

1.1 Background

Bangladesh Agriculture is highly dependent on seasonal climate, the soil condition and production pattern of a district. Crop decisions are often made under uncertainty: what seemed like a good land by experience, the final crop performance can be influenced by rainfall, temperature, vegetation condition, soil chemistry, and past productivity of the district. This uncertainty is particularly critical to Aman rice and wheat, two crops that reflect various seasonal and agronomic needs.

The recent advances in machine learning have provided viable prospects to agricultural decision support [14], [15], [22]. Remote sensing can summarize vegetation and surface states [34]; climate services can give temperature, rainfall and solar radiation indicators [37]; gridded products like SoilGrids can give soil property priors [35] and computer vision models can classify soil appearance based on field images [7], [32]. But these sources of information are usually used independently. A yield model can disregard visual soil evidence, whereas an image classifier can not take into account the district-level crop performance.

This study creates a multimodal machine learning framework to integrate the soil images classification, tabular yield prediction, and rule-based crop-soil suitability

to support sustainable crop decision-making in Bangladesh. The architecture is geared towards decision-level fusion as opposed to end-to-end neural fusion. A soil photograph fits into one of seven soil classes and one flood image class, district and seasonal data are utilized to predict Aman and wheat yield, and transparent agronomic guidelines sift crops that cannot be cultivated in the predicted soil type. The project is not aimed at making a field-validated agricultural advisory product. Rather, the thesis illustrates a demonstrable engineering prototype that has the potential to bind together several data modalities into a feasible recommendation process. This scope is significant as paired soil-image and field-yield ground truth are not provided in the existing dataset. Thus, the system is tested at the component-level and by an ablation of a proxy fusion.

1.2 Rationale of the Study or Motivation

Even though single-modal assessment techniques often yield valuable results, they often give a limited view of agricultural systems. In relation, a multimodal framework, which is an integration of geospatial imagery coupled with time-series climate records and socio-economic variables, gives a more realistic and all-inclusive evaluation. Of special significance to such an approach are the smallholder farmers and policy planners who need decision-supporting mechanisms that are precise, interpretable, and scalable.

Past research has shown that multimodal frameworks have achieved greater predictive success and improved generalization across healthcare, autonomous systems, and other domains [27], [29]. Similar results are expected to be realized in the area of agriculture, where modalities that have traditionally operated independently are merged. This ability of any multimodal machine-learning method to contribute to sustainable agriculture, particularly by improving crop yields and reducing environmental damage, is an adequate explanation of the current research.

The underlying issue of agriculture has always been the effectiveness of information and how to utilize the weather, soil, and other information to maximize crop production while making a sustainable crop cycle. Especially in a lesser developed country like Bangladesh, the absence of this applied technology is costing the farmers and the people dependent on agriculture, missing out on a very crucial opportunity to not only expand their financial gains, but also expand the whole production to a level that the export market has an uprising. This is integral to building our economy and growing our GDP and satisfying the hunger of this country's ever-growing population. The research aims to provide a robust solution to make this technology accessible and usable to direct the agricultural structure of Bangladesh towards a more efficient, organic, and sustainable production cycle.

1.3 Problem Statement

Extension stakeholders and farmers require crop recommendations that are based on more than just one signal. Even when the soil type is unfavorable, a yield forecast can suggest growing a crop that yields highly. A soil classifier can only determine soil taxonomy, but not the anticipated crop yield of a specific district and season.

Moreover, there are numerous tools of agriculture, which are data-driven and hard-to-interpret since the decision logic is concealed within one model.

The main issue that this thesis will address is how to integrate evidence of soil images, district-based tabular yield projections, and crop-soil suitability regulations into a clear recommendation system of Aman rice and wheat in Bangladesh. The system should also be transparent regarding uncertainty particularly where there is low confidence in soil classification or where there is no actual field level evaluation data.

1.4 Research Objectives

Multimodal Soil Imagery Classification framework: To develop, train and test a deep Convolutional Neural Network (CNN) optimal architecture for fine-grained classification consisting of eight distinct soil classes. A stratified sample pipeline will be employed to split the dataset into non-overlapping train, validation, and test subsets, ensuring that class distributions are maintained, training set imbalance is avoided, and generalization metrics through blind testing are produced in an unbiased fashion.

Predictive Fertilisation for Resilience and Efficiency of Cultivar Systems: To develop resilient data-driven crop models for development of robust and predictive fertiliser systems in Bangladesh specifically for Aman rice (BSFC) and wheat. This entails melding à la heterogeneous, high-dimensional feature space which fuses multi-spectral remote sensing indices, multi-temporal historical climate profiles, gridded physical and chemical soil attributes (e.g., nitrogen concentrations), geographic district indicators as well longitudinal historical crop yield statistics — to identify complex mappings from environmental drivers to productivity outputs.

Decision-Level Multimodal Fusion & Yield Prediction Policy: Implement an algorithmic decision-level fusion policy to mathematically couple the decoupled outputs of the vision-based soil CNN classifier into tabular categorical and regression class structures. The completed framework will generate accurate and context-aware final crop yield predictions by dynamically merging high-level geospatial priors with ground-level visual evidences to disambiguate conflicting modality signals. Challenge our architectures against quantitative standards of soil classification performance and yield predictions by setting the bar Empirical Validation, Structural Ablation, Regional Exploratory Diagnostics This diagnostic pipeline will merge temporal yield holdout validation to test model robustness to future climatic variance, focused regional exploratory analysis of accountability to soil, biomass and yield via GEO observation and WAN networks, and structural ablation studies to isolate, quantify and test the empirical effect of the multimodal fusion layer in a rigorous way.

Explainable User Interface & Streamlit Deployment: We created a production panel-ready web app with the use of streamlit focusing on a use-case in an interactive web-app that creates an intuitive user-friendly interface for end users, agricultural extension officer, and policy planner. The interface will expose local crop recommendations and quantitative prediction uncertainty intervals, and an explainable artificial intelligence (XAI) layer will justify all system outputs using auditable transparency rationales.

1.5 Methodology in Brief

The research methodology will involve the following key steps:

Figure 1.1 summarizes the end-to-end thesis workflow from background study through evaluation and future scope.

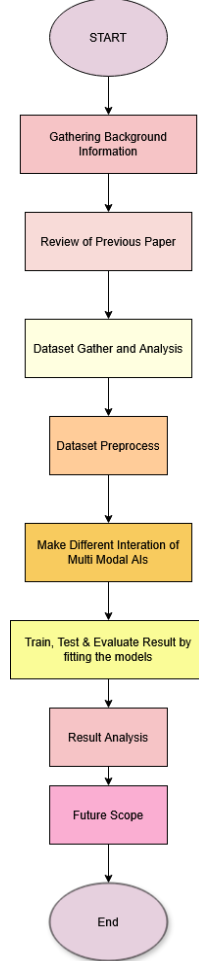


Figure 1.1: Multimodal Learning Framework Flowchart. Sequential stages from data collection to model evaluation and reporting.

Collecting the background information: familiarize with the fields of sustainable farming and machine learning fundamentals.

Analysis of past articles: conduct research on past studies involving unimodal and multimodal AI applications in agriculture and other areas.

Data collection and analysis: find, obtain, and develop geospatial, climatic and socio-economic data.

Preprocess dataset: clean, normalise, and prepare data for training; deal with missing or inconsistent values.

Develop alternative multimodal models by making use of fusion (early, late, hybrid).

Train, test and evaluate models using performance measures (e.g., accuracy, RM-SLE, precision, F1-score) and compare fusion strategies.

Future scope: recommend potential enhancements, practical relevance and future directions.

1.6 Scope of the Study

The architecture scope of this framework is explicitly characterized by the hand-picked composition, geographical scaling and spatiotemporal evaluation strategy of its multimodal dataset. The integrated image repository formed by the data from various sources is specialized to eight specific environmental profiles—seven different soil taxonomic classes (Alluvial, Arid, Black, Laterite, Mountain, Red and Yellow) and a categorical class for flooded land imagery. This computer vision pipeline is placed in dynamic with tabular predictive models whose functional application will be purposefully restricted to furthering the recommendation and optimization of two core strategy crops in Bangladesh—Aman rice and wheat. Instead of being based on precise and incessantly available GPS coordinates — which often are not available, not reliably logged or hard to generalize over different clusters of smallholder farming — this yield forecasting infrastructure is logically constructed solely as a series of interlinked but localized district-level models. Thus, the user explicitly specifies an administrative district during inference so that predictions respect macro-regional dynamics. To assess the robustness of the framework under more forward-looking environmental changes, we adopted a strict temporal holdout validation procedure where all predictive models are trained sequentially on previous chronological records and evaluated aggressively against an entirely unobserved future test year.

On the methodological side, the framework deliberately steers clear of the computational and optimization complexities imposed by end-to-end multimodal deep learning, favoring an aggregated decision-level fusion paradigm optimized for small-scale research datasets. **Structural Policy:** This policy deliberately avoids simple concatenation of high-dimensional image embeddings directly with tabular feature vectors as was historically done, nor does it attempt the highly erratic challenge of predicting quantitative laboratory soil chemical properties such as nitrogen or pH from raw pixel values. The vision-based soil classifier is instead used as a stand alone, specialized module, where the output(predicted taxons) are only used for determining soil taxonomy. This taxonomic determination is then routed by a deterministic, rule-based agronomics suitability layer where it is combined with the independent yield predictions from the tabular models. The framework circumvents the well-known serious risks of overfitting and representation collapse that can occur while training end-to-end trained deep multimodal networks with limited data by treating the system in the form of a decision-level fusion pipeline instead of a complicated "black-box" neural architecture. Additionally, this modular abstraction offers a very natural and clear route to explainable artificial intelligence (XAI), thereby substantially increasing the safety and interpretability of its final recommendations for agricultural extension officers and end-users.

1.7 Contributions

- A reproducible soil classification benchmark for diverse architectures including EfficientNet-B0, MobileNetV3-Small, ResNet18, ResNet34, and a shallow CNN.
- Crop-to-crop cross-validation experiments of an XGBoost yield forecasting workflow with temporal holdout evaluation, using temporal holdout backtest-

ing protocols for Aman and wheat.

- A fusion policy that purges poor crops, ranks potential crop species by median yield and abstains with low soil confidence.
- A small set of the research artifact, including the thesis tables and figures, ablation results, and a working GUI demonstration.

Chapter 2

Literature Review

The chapter is a complete overview of research and basic approaches on the current state of the art of the fields of multimodal machine learning, computer vision and data-driven precision agriculture. Traditionally, computational tasks making up smart farming workflows have been deployed in unimodal silos, with one working on leaf taxonomy by computer vision, another on macro-regional climatic trends by tree-based ensembles, and a third based on deterministic agronomic rules. Individual models perform well in their respective areas, but are by nature unable to represent the intermodal relationships and localized idiosyncrasies of a complex agricultural system in deep uncertainty. This review critically exposes a key operational issue; the absence of a clearly defined, resource-efficient framework to align ground-level visual evidence with tabular spatiotemporal priors, while systematically surveying early, late, and hybrid fusion topologies, as well as foundational transformer models, edge-deployed robotic sensor networks, and deep convolutional architectures (CNNs). Thus, this survey provides empirical and theoretical support for the prototype of modular late fusion developed in this study, which serves as an essential barrier to maintaining the stability of a model and offers interpretable and risk-aware crop recommendations for Aman rice and wheat in the particular environmental context of Bangladesh.

2.1 Preliminaries

Multimodal machine learning is a set of techniques aimed at receiving, mapping and fusing multiple heterogeneous data streams to obtain a comprehensive assessment of complicated systems. Specifically, within this agricultural construct the generic representatives are classified into three unique environmental dimensions;

Geospatial and Reconnaissanced visual features Ground-level field photographs as well as segmented environmental frames can be the main input of deep Convolutional Neural Networks (CNNs) to predict local surface properties and classified taxonomic information at a fine-scales over large areas in disparate landscapes.

Categorization of Climatic parameters and environmental variables: Multi-temporal time-series records (i.e., rainfall parameters, temperature dynamics, as well as solar radiation indicators) are estimated through tree-based ensemble architectures when linked with retrospective agricultural functions to facilitate extrapolation over a temporal range in conjunction with historical anthropogenic influences.

Physiographic and Administrative Variables: High dimensional tabular datasets

containing aggregated gridded soil properties (e.g. mean regional pH, clay fractions, nitrogen concentrations) combined with categorical macro-regional administrative district identifiers that govern geographic crop zoning

Contemporary multimodal architectures interact with various datasets by organizing fusion topologies into different paradigms where the points of intersection between streams of information are:

Early Fusion (Feature-Level): The concatenation or mathematical combination of the original or intermediate feature vectors extracted from independent modalities which can then be fed into a single joint learning network.

Late Fusion (Decision-Level): a cautious modality fusion setting with domain-specific models separately processing their respective modalities in full isolation. These distinct output vectors (like independent probabilities for computer vision classes and tabular regression predictions) are then aggregated via rules, voting heuristics or scripted policies to produce a single final recommendation.

Hybrid Fusion: A two-phase method that combines both low-level feature fusions (early fusion) and high-level structural ensemble methods (late fusion) in the same model pipeline.

By utilizing cross-modal logic automation, these modalities, based on networks, allow machine learning frameworks to transcend the limitations of single-modal judgment methods. Module decision-level late fusionomics is an empirically valid guardrail for research datasets that are either constrained datasets or unpaired datasets, for which end-to-end alignment is impossible. Thus, providing a shield over end-users from potentially harmful recommendations and training logics with highly interpretable decision-making preserving structure stability against representation collapse.

2.2 Review of Existing Research

In the paper entitled “AI Beyond Text: Unifying Vision, Audio, and Language for Multimodal Learning,” they explain the fundamental limitation of single-modality artificial intelligence machinery, relying strictly on one of the sensory perceptions, such as sound perception, visual perception, or text. These techniques confine their parsing of low fidelity, real-world situations in that a textual format such as a sentence, cannot handle the emotional tone found in pure speech, likewise, such a system would struggle to capture text-based contexts like those found in speech. As a consequence, this conversation identifies an important consideration for building complete AI systems capable of simultaneously synthesizing and processing information from different sensory modalities To unparalleled levels of interpretation and success in complex applications.[29]

Raising the question of the necessity of frameworks which would allow exploration of the synergistic potential of many perceptual modalities, the current study answers this question in the affirmative. In that regard, it discusses three modalities such as the vision, audio, and language and uses the analytical tool of computer vision, speech recognition, and natural language processing respectively. The piece is an extension of the previous research in the field of modality fusion and provides a detailed presentation of three modalities paradigmatic approaches, that is Early Fusion, Late Fusion, and Hybrid Fusion. The Early Fusion works by combining the feature inputs of both modes before the internal processing takes place, hence

necessitating a unified representation. Late Fusion on the other hand uses each modality separately and then combines the resultant representations. Hybrid Fusion combines the methods by utilizing early and late fusion performance on lower and higher level processing respectively. This is followed by a comprehensive exploration of deep neural networks; discussing Convolutional Neural Networks (CNNs) that are used to process images, Recurrent Neural Networks (RNNs) that are used in processing sequential data and transformers that are used in attention between modalities.[27] In this method of analysis, the research draws the line of systems that would be able to process visualizing systems generated by cameras and medical image sets, audio input speech and the external noise, and linguistic systems of text-based documentation and patient files.

Multi-sensory approach was confirmed to be an efficient tool of improving the functioning of AI in a vast variety of categories. In the healthcare sector, the research shows that multimodal systems (the combination of medical imaging and electronics containing patient records together with verbal notifications) delivered improved diagnostic accuracy. An example is Google diabetic retinopathy diagnosis algorithm, which stated that an increased level of diagnostic accuracy was observed when combining retinal images with patients having their previous history. Multimodal approaches performed better across tasks that needed a thorough comprehension compared to the single-modality approaches. Findings indicate that multimodal systems cut down error prevailing at 15 to 30 % of various implementations compared with the unimodal alternatives.[26]

Multimodal approaches performed better across tasks that needed a thorough comprehension compared to the single-modality approaches. Findings indicate that multimodal systems cut down error prevailing at 15 to 30 % of various implementations compared with the unimodal alternatives.[26]

The current study gives a critical analysis of the transformative potential of multimodal AI systems. The key point of information is the difference in the output of three types of fusion, early, late, and mixed ones, in accordance with the target application. An intensive analysis of practical implementation in healthcare, autonomous systems and customer services provides answers not only to the apparent limitations, but to methods to cope with them as well. Of particular interest is how focusing processes work in transformer-based architectures in cross-modal learning, that is, allowing us to draw dynamic focus on meaningful information across complementary modalities.[29] Multimodal system design involves a complicated technical issue of data alignment addressed in the discussion of the issue of parallel temporal and spatial synchrony of heterogeneous data.

The other glaring omission in the above paper is the issue of computational complexity and real-time data constraints. [12] The paper presents an excellent representation of fusion schemes and uses, but it fails to give a proper evaluation of computational overhead, which is embodied in concurrent processing of data streams of high dimensions. In addition, the issue of scalability in multimodal systems, especially in resource-limited contexts like mobile devices and edge systems, has not been explored fully.

Another weakness is in the fact that it has given little coverage to the management of missing or corrupted modality, a common occurrence where a single or multiple input streams have briefly become unavailable or disabled. It also does not pay enough attention to a related issue, the imbalance between modalities, a related issue that

various types of input may imply different information content in reliability. The current work of the research is aimed at alleviation by creating adaptive multimodal models that could alternate processing strategies depending on the availability of resources and quality of incoming data. The pursuit of this is exploring lightweight fusion procedures that maintain performance and reduce computational intensity, methods like knowledge distillation and pruning explicitly developed under multimodal circumstances.

Such a comprehensive review establishes the basis of some fruitful areas of further investigation. To begin with, the constructing field of neuromorphic computing can, in principle, address the computational efficiency limitations, revealed in the research. Second, investigation of federated learning on multimodal contexts would facilitate privacy-preserving learning on disaggregated datasets as well as the rich representational capacity explained throughout the text. As a whole, the paper forms a strong basis on which further investigations on these advanced branches could be made, and its methodology which reflects the procedure of larger evaluation of multimodal applications of AI is an effective template that could be used to evaluate future principles of this swiftly developing field.

A pertinent weakness to traditional agricultural research (referenced in the current report) is a demand on numerical-based information that is produced during field and laboratory research exercises. Such a state of affairs has brought quite significant rewards, but is more likely to fail to recognize other streams of information that can be just as valuable, such as visual data (i.e. photographs of pests) and unstructured language data (i.e. field reports). Such multimodal data when combined can provide more valuable information about agricultural phenomena such as managing pests as well as providing nutrients in an equitable manner.

Another related issue is the transboundary pest detection. It used to require specialist skills and tedious inspection of the field and was not only time-consuming but also subject to error. Compared to that, using highly advanced processes, predictably artificial intelligence, it is possible to automate the combination of multimodal data inputs, like satellite imagery, weather data, and previous research findings, thus enhancing the precision and quickness of detection. An analogue is the delivery of personalized nutrition (i.e., dietary advice depending on personal health profiles). Where the current practice tends to be all-inclusive in its recommendations, modern artificial-intelligence algorithms, informed by large databases of food intake, genome, and physiological measures, can prescribe a diet that is, in comparison, exceptionally finely tuned. The current article explores the possibilities of multimodal artificial intelligence (AI) in solving issues of the agricultural sector. Multimodal AI brings harmony between heterogeneous modalities of analysis, such as text, imagery, and sounds. Two areas that its use promises to be especially fruitful are outlined in the following sections: Agricultural Text Natural Language Processing (NLP). The authors emphasize the possibility of NLP to transform the vast body of output of agricultural research into applied knowledge. The National Agricultural Library (NAL) of USDA is already using NLP methods to index and categorize agricultural literature automatically. It helps the researchers to better find the pertinent information faster. The efficiency increase of indexing brought about by this automated process has, in practice multiplied by seven. [22]

Pest Control and Crop Monitoring Computer Vision.[30] The report further shows the potential of computer vision in the identification and surveillance of crops, pests,

and diseases on the basis of image analysis. The method is more relevant in the cultivation of high-value fruits; such manual processes as the detection of flowers, monitoring of pests, and fruit selection remain labour-intensive and are carried out with the aid of human eyes. AI systems are capable of analysing pictures of flowers, pests and fruit within a short time and with high accuracy by using deep learning models.[17] As an example, the paper describes a deep learning model that is presented to detect flowers in apple orchards and that outperforms manual solutions especially due to changing lighting conditions. Lastly, the researchers examine the aspect of multimodal data integration into personalised nutrition. AI may take into consideration meal photos, health databases, genetic or microbiome data to provide more personal nutrition advice in this area. According to them, such integration of data would provide more applicable and accurate nutritional advice, and thus a better health outcome. Collectively, they demonstrate a strategy involving use of multimodal data streams and AI methods to address the different agricultural challenges innovatively. Through concomitant text and image, text and picture, and other modalities analysis, researchers can draw deeper conclusions and invent better interventions.

The empirical research shows that the multimodal AI approaches have produced positive results in various areas of farming. The use of image identification combined with sound analysis in detecting pests shows a greater level of precision and efficiency in the effectiveness of detecting invasive species entering a country through its border inspection points, thus reducing the financial impact of being exposed to damages through these pests. Personalized nutrition: In personalized nutrition, in comparison with traditional approaches, models incorporating both food intake information and other health data based on AI are more accurate and effective to enable custom and efficacious diet interventions. With a direct comparative analysis between multimodal AI implementations and the single-modality implementations of conventional AI, it is possible to conclude that the multimodal setups will make a considerable difference in terms of diagnostic accuracy improvement and the overall increase in the efficiency of operation in pest detection. Correspondingly, in dietary personalised nutrition, the combination of AI models is more effective than individual sources of information, like food diaries or self-reports of the consumption, since multi-modal models can provide a more thorough picture of the diet and be used as an objective source of dietary data. In general, the results clearly show that multimodal systems tend to outperform unimodal alternative in answering complex agricultural questions because of the different capabilities of the individual data modalities, such as exploiting the strengths of the different data modalities and having the ability to produce more detailed information and contribute to a significant reduction in the error rate.[18]

In the context of purely academic research, AI can be successfully applied in such practices as automated scholarly literature indexing and automated pest detection, meaning that the scope of human time consumption on such efforts, as well as the cognitive burden is largely alleviated, allowing researchers to concentrate their efforts on more strategically minded research.[16] Moreover, the systematic integration of AI products into everyday farming practice provides the opportunities to improve sustainability, reduce the labour costs, as well as increase the crop yields by adopting data-driven informed decisions.

Knowledge about methodological hints that could be applied in the studies con-

ducted by researchers that are working at any area where multimodal data are made on a comprehensive level are also provided in the study. As an example, analysis methods like text analysis using natural-language processing and image classification using computer vision are easy to transfer and then adapt to similar tasks in the fields of healthcare, autonomous systems, and metropolitan planning.

Despite the fact that the results of this study are promising, a variety of limitations and improvement opportunities are noted in the sphere of multimodal AI in agriculture. First, the lack of high-quality data can be considered a significant challenge, especially when it comes to training AI models. For example, in pest detection, the availability of annotated imagery on pests hinders building powerful systems. Similarly, personal nutrition apps must rely on very large, high-quality datasets so they can make confident estimates of how diet will affect health.

Second, in the paper, another limitation is mentioned in terms of differences in the level of expertise in machine learning between researchers and industry professionals in the field of agriculture. The authors allege that not many practitioners possess the technical skills that are required to efficiently apply AI approaches. There are already USDA initiatives that are aimed at developing a competent workforce that would be able to utilize such techniques.

Third, the combination of heterogeneous data, even when data come out of distinct sources, formats, and time frames, is always a challenge. Continued development of AI systems to process and interpret this multimodal information on a real time basis is important.

The article by Uchechukwu C. Ajuzieogu of the University of Nigeria titled Multimodal Generative AI for Climate Adaptation and Agricultural Innovation is ambitious and visionary as the author provides a comprehensive outlook on the whole complex of the issue of climate change in global agriculture and proposes a multimodal generative AI approach to resolving this problem. The article has a technical soundness and socio-environmental topicality.

The key finding in the study by Ajuzieogu is presented, to a large extent, in a multimodal AI structure that integrates information found in a mismatched set of sources, such as satellite photos, ground sensors, historical data on yields, market indicators, and folk wisdom. Intended to aggregate an information dense view into usable, location-specific agricultural suggestions, the framework is particularly sensitive to the smallholder farmers who are most exposed to climate shocks but have least access to high-tech instruments.[28]

The study concerns a major contemporary issue affecting the world as a whole: the growing gap between the immense data generated by modern climate science, and the ability of agricultural civilization, especially smallholders, to use this data in making practical, short-term decisions. Farmers are working in a more complex environment where their traditional agroecological knowledge is rendered insufficient by the rapidly changing climatic trends, whereas, most of their AI systems and technical tools are too generic, too resource-demanding, or unapplicable to the rural environment with its poor infrastructure. This challenge is viewed by the study as a lack of accessibility, contextualization, and integration.[28] Though raw information is mostly easily accessible, specific knowledge that is both scientifically precise and culturally relevant is considerably lacking. Also, farmers are often forced to choose between piecemeal recommendations, mostly generic smartphone apps, or antique calendars unable to adapt to the current climate variability. Its essence hence is not

that there exists insufficient data but the failure to transform the information into usable knowledge locally.

The framework developed by Ajuzieogu is not only a predictive model, it is a complete end-to-end AI ecosystem that was iteratively developed, field-tested and refined over several years, in a partnership with 47 farming communities in 12 countries. One can sum up its operation sequence as follows:

The system incorporates a wide list of data feeds: Satellite images of high resolution by Sentinel-2, Landsat-8, and Planet Labs. Soil and weather sensor matrices in the short intervals. Historical and seasonal crop data.[14] Climate outlooks and live environmental factors Traditional and aboriginal, ethnographically recorded knowledge.

Upon data synchronization, in the temporal and spatial dimensions, a sophisticated cross-attention architecture is put in place. The generation of precise, time accurate and spatially aligned insights is based upon the dynamic time- warping methods and multi-scale feature extraction.

Conditional generative machinery enable generation of: A soil health-based and climate stress-based soil health based crop selection strategy, driven by market tendency. Guidelines on the use of resources such as irrigation and provision of nutrients. Touch screen alerting, in case of extreme weather conditions and pest outbreaks. The time of planting and harvesting according to the micro-climate forecasts.

Ajuzieogu uses transfer learning to customize core models to the local environment, cutting training by 67 % and increasing adoption chances.[28] Its implementation architecture is modular and scalable since it can be deployed in low-bandwidth conditions, to mobile devices with limited processing resources, multilingual and voice-based interfaces.

The result was a system that was able to perform well even in real-world, varied farming environments. The accuracy of yield prediction was achieved up to 85% which helped farmers in taking decisions on time and with data. Water usage dropped by up to 42%, thanks to AI-optimized irrigation schedules. Climate risk forecasting—covering droughts, pests, and extreme weather—proved highly effective, giving farmers early warnings and improving crop survival rates during stress conditions.

Compared to traditional advisory methods and unimodal AI tools, this multimodal approach outperformed in adaptability, precision, and farmer usability. It wasn't just technically sound—it was usable, scalable, and embraced by local communities, with adoption rates exceeding 75%. Unlike static models, it continuously learns from feedback, proving more resilient and relevant over time.[9]

Lastly, the system is expected to be edge-first; with federated learning and peer-to-peer data synchronisation mechanisms to keep the reliance on technologies associated with the cloud to a minimum, and, thus, minimise the latency.

Preliminary studies prove that multimodal information ought to be procured collectively during early fusing stages or consolidated some time after modal scrutinizing in dwelling fusion. Besides, the current paper offers a prospective common denominator of systems that could derive meaningful conclusions about the combination and transformation of perfect or incomplete data and as a result, solve the problem of reliability and shrinking dimensions. Lastly, the scope covers to describe the processes which will enhance reliability of trust and transparency within AI-generated advice as the primary concern lies in social acceptance and governance.[28]. Simul-

taneously, a multi-objective optimization approach is used to calculate economically viable, sustainable solutions, which further the national policy goals.

Overall, the results indicate the need to re-contextualize AI performance metrics to include equity and resilience, in addition to performance, therefore providing a guide to practitioners whose mission is to engage in a responsible and fair technological growth.

Despite a number of limitations, the current study is a good contribution to the body of knowledge. **Constrained Computation.** However, the results prove quite accurate and the inference time is low, whereas the current training and model frameworks equate to large computational requirements. Most agricultural producers use entrylevel smartphones, and, therefore, the future study should specify the best balance between utilisation of the resources and the model efficiency, i.e. the consumption of memory and energy consumption.[9]

Availability and the Quality of Data. The model is sensitive to the completeness and consistency of the input, and all too often a sensor fails or the line-of-sight situation is intermittent and spikes in historical records are common situations facing in an operational environment. It has to be extensively ascertained that it works when some of the data or some of the model is not known.

Interoperability, and standardisation. The authors note that the suggested structure can be combined with the existing agricultural systems, but since the interfacing mechanisms with existing systems is discussed rather superficially, inclusion of more specific information of that kind would add to the piece. Large-scale deployment could be made affordable because of standardisation protocols and interoperability with national databases, or IoT infrastructures would contribute to the delivery of large-scale deployment.

The Social Learning Metrics. The system promotes decentralisation of feedback and suggestions, but the research is not used to show how AI suggestions have a long-term impact on changing the behaviour of farmers and exchanging information among villages. **Uncertainty Analysis on Recommendation systems** Climate modelling enabled by quantum Quantum-enabled impact of climate on the arctic environment **Supply-chain traceability with support of blockchain** Connecting to the supply chain **Spreading of ecosystem services metrics to the categories of biodiversity and soil health.**

The paper by the author Uchechukwu C. Ajuzieogu called Deep-learning-based specifications in 2018 is a work of seminal value that proves the ability of deep-learning methods to go far beyond the sphere of high-income laboratories and commercial implementations. By quality of the very design of a system sensitive to the needs of farmers, to the movement of local civilizations, and to demands of a global ecology, Ajuzieogu creates architecture that is both morally sound and technically refined.

Teng et al., published a recent article in 2023, in which they proposed a systematic multimodal sensory data, which targets agricultural robotics on citrus orchards. With Hanzhe Teng, Yipeng Wang, Xiaao Song and Konstantinos Karydis as leaders, the group combined the citrusalarming dataset to develop autonomous robotics in precision agriculture combining sensor fusion and mapping techniques.

The article fills in a vital loophole in contemporary research on agricultural robots on the account of highlighting the lack of extensive datasets which incorporate multispectral imaging and navigation sensor modalities altogether. The typical practice of monitoring crops is that it is labour-intensive, time-consuming, and cannot be

expanded geographically and (temporally) across spans of time [10]. The authors note that existing datasets available publicly provide access to rather limited sensing bandwidths - usually 3-6 modalities - that numerous applications in agricultural robotics would require at higher frequencies to assist in precise localization, mapping, and crop-monitoring tasks. In such a way, the study illustrates the importance of having strong datasets that allow the creation of self-governing robotic systems that could be used to conduct regular and automated observation of the condition of the crop in the controlled agricultural premises.

The paper has implemented a robust dataset development process on a wheeled mobile robot platform Clearpath Jackal with nine different sensing modalities. The methodology of data collection will be composed of the components in the following way:

The mobile system will use the multiple sensors such as the FLIR Blackfly monochrome camera, the Stereolabs Zed2i stereo RGB camera, with depth computation, the Mapir Survey3 Red-Green-NIR camera that calculates vegetation index, the FLIR ADK thermal camera, the Velodyne VLP-16 3D LiDAR, the MicroStrain IMU, the wheel odometers, and the SwiftNav Duro GPS-RTK system to ensure centimeter-grade exactness.

Three steps calibration process were followed: (1) calibration of multi camera intrinsic and extrinsic parameters with the Kalibr toolbox; (2) IMU-camera extrinsics; and (3) LiDAR-camera calibration with the ACFR toolbox. The most prominent advances are removing possible thermal camera calibration errors under direct sunlight conditions to allow the recognition of the checkerboard pattern. Three citrus fields of varying nature were examined with seven sequences used: sparse uniform planting (Field One), semi-dense in-row planting (Field Two) and dense in-row planting (Field Three). The collection included various stages of growth, planting schemas, daylight conditions as well as movement patterns which comprised the patterns of lawn-mower, square-spiral and every-other-row movement. CitrusFarm dataset is thus a big step compared to other agricultural robotics datasets especially when considering the issue of sensing modality variety and completeness of the data. Its main quantitative characteristics are concluded as below: which 1.3 TB of multi-modal data could be obtained. The respective sequences are flanked by synchronized data streams at optimal temporal resolutions: 10 Hz by cameras and LiDAR, 200 Hz by IMU as well as 50 Hz by wheel odometry.

In line with the rest of the literature, when beyond the context of analogous studies (Chebrolu et al., 2017; Bender et al., 2020; Polvara et al., 2023), CitrusFarm is unique as it can record nine classes of sensing modalities at a time, with the previous approaches typically providing four to six modalities. This contribution is distinguished by the existence of thermal imaging, multispectral NIR options and ultra-precise GPS-RTK positioning.

The manuscript presents evidence that shows that calculating agricultural indices that are specific to the agricultural quadrant, like NDVI based on the channels Red-Green-NIR, is possible to directly assess the condition of the vegetation. The accuracy of the GPS-RTK system was preserved at the centimeter level during the data collection session, which permitted having a homogeneous ground truth to use in further localization procedures. This study implies several breakthroughs in agricultural robotics since it suggests the following developments: Multimodal Sensor Fusion: The gathered data allows analyzing higher-level sensorfusion techniques

incorporating visual, thermal, spectral, and spatial sensors so that the developed agricultural robot operating on the light-varying pathways can work effectively.

Agricultural Applications: The software produced on the multispectral imagery gives a real-time access of the health of the vegetation through calculation of NDVI, presence of water-stress through thermal-image analysis and fine mapping of crops using the RGB multi depth information at high resolution. **Localization and Mapping Improvement:** The combination of GPS-RTK, stereo vision, IMU, and LiDAR offers a dense ground-truth reference against which SLAM calculations can improve those soldered to agriculture, where visually inarguable and repetitive patterns and the occasional and predictable loss of the GPS signal within tree canopy are common problems.

Deep-learning Models: The large scale data size of 1.3 TB could be used in training the deep-learning models on complex tasks like detection of crops, classification of growth-stages, diagnosing healthy plants and autonomous navigation in the controlled agricultural settings. Even though the given dataset is rather massive, there exist numerous restraints in terms of scholarly enhancement: environmental diversity, dynamic content, temporal coverage, ground-truth annotations, processing framework, and data integration.

To begin, the dataset is limited to citrus orchards of one geographical location at a time of the year (summer), thus limiting the transferability of the results to other crops, seasonal and climatic variations. Secondly, the set of data overlooks dynamic factors, which are common in the real-world farming practices, i.e., agricultural machinery motion, livestock transportation, and changing weather conditions in the data-gathering process. Thirdly, the data has no longitudinal temporal coverage, despite the use of a number of points along the diurnal cycle, and this would not be suitable in phenological studies and in monitoring crops over time. Fourthly, the dataset provides sensor calibrations and positional ground truths, but not the widespread semantic annotations with respect to crop health status, presence of certain disease, or counts on amount of fruits, etc. such annotations which would have taken it even deeper into the realm of supervised learning. Consequently, the holistic sensory data lack conventional metrics of evaluation and procedure of benchmarking that is typical of an agro-robotics operation, which hinders comparative studies. Finally, 1.3 TB of data and its heterogeneous format is also computationally hard and difficult to store, so compressed or processed derivative data are to be considered.

The ability to harmonise disparate data has become an important aspect in research that tries to focus on sustainable land governance. Combining geospatial images with socio-economic indicators, climatic time series, etc. can make the selection of land use and resources allocation aspects more precise. In order to support this integration we analyze two surveys on multimodal learning approaches. The article by Sundar and Heck (2022) determines the current status in the field of multimodal conversational AI, focusing on collections and approaches to integrating visual, linguistic, and other modalities into dialog systems. Equally, Peng Xu et al., (2023) provide a taxonomy (based on transformer) of vision, language, audio, and other data types broadly under a single framework. The next part summarises what each of the studies investigates, the methodology used, the aims of the studies, and highlights of findings and then compares the two papers in terms of the scope of their study, and the direction taken. Also, we address each of their contributions to the proposed agricultural use case and outline their complementary limitations. Last

but not least, we place our thesis into the context of the larger body of literature by building on the knowledge of these surveys and uniting the gaps they expose.

In the 2022 NLP4ConvAI workshop paper Multimodal Conversational AI: A Survey of Datasets and Approaches, Anirudh S. Sundar and Larry Heck of Georgia Tech engage in a methodical survey of how far to extend the conversational agent beyond text to the whole range of human senses. They base their work on years of experience in the field of dialogue and conversational interface. In this context, the authors define the main goal of multimodal conversational AI in mathematical terms: an agent should produce proper answers to the inputs of various modalities. To deal with this goal, they break down the problem into various sub-problems representation, fusion, translation, alignment, and co-learning each one of them is a step towards the process of encoding modalities, integrating modalities and applying the resulting representations to reason across modalities. In the table offered by Sundar and Heck, their taxonomy is summarized in a concise manner.

The authors then focus on the state of the art and review the recent developments in every element of their taxonomy. Such types of representation and fusion (such as joint and coordinated embedding, which incorporates information on text, visual data, and audio) are represented. Same is done with translation and alignment with emphasis on models that anchor language in a viewable or gesture modality. The talk is Deliberated with a rather comprehensive listing of multimodal dialog and QA datasets (VQA, GuessWhat, SIMMC) and vision-language benchmarks. In their qualitative evaluation of these corpora, one factor stands out, a discrepancy that has been noted before: most of the datasets involve visual question-answering or single turn-based tasks, whereas most of the interesting dynamics and complexity of ordinary conversation have not been captured. Specifically, image-based QA dataset focuses on clear questions regarding the images, but not on the more covert signals and interactive nature of the conversation that make this dialog interesting. The video and action datasets, e.g. UCF101 and Kinetics, provide language annotations but not the dialog scene that is a part and parcel of the multimodal dialogue.

The lay-off in power among benchmark constraints is one of the recurring themes of the survey by Sundar and Heck. As an example, the Image-Chat dataset allows visual conversation on multiturn interaction and is restricted to three sessions and image and text. With such layouts, argue the authors, it is not surprising that the best models including TransResNetRet, tend to hallucinate or to fail after a couple of exchanges, as they were not trained on dialogues of a deeper or a wider nature. To conclude, one could say that achieving the full potential of multimodal dialogue will necessitate new datasets, which is representative of the realistic conversational dynamics, as well as new model architectures that will be able to learn a single consistent objective among all accessible modalities.

However, we have to bend when shifting the conversational orientation that the survey had to the field of agriculture. The data used in the analysis of Sundar and Heck is virtually urban/household-specific, thus leaving untapped spatiotemporal assumptions that characterize land management. The models considered are strongly dependent on visual and textual data; geospatial location coordinates or time-based measurements (which are vital in agronomy decision making) are nowhere to be found. Lastly, the authors admit that the current methods are optimized on small tasks (image captioning or QA); performance may be seriously hampered by a similar limitation in the agricultural sector. As result, our thesis will expand the framework

Sundar and Heck in some directions: concatenation of more modalities (continuous climate streams), addition of more link in dialogic chain, longer and more involved reasoning tasks, and a training of models under a single set of objectives that match with real-world agricultural objectives.

The second poll is Survey on multimodal learning with transformers by Peng Xu (Tsinghua University), Xiatian Zhu (Univ. of Surrey) and David A. Clifton (Univ. of Oxford). This paper, currently under acceptance in IEEE TPAMI, adopts a complementary view instead: it takes an overview of the Transformer architectures that support current multimodal learning in many fields. The motivation of the authors to conduct the survey was the emerging abundance of multimodal Transformer techniques and the associated necessity to obtain a comprehensive structured overview of the achievements. They suggest a 2-dimensional taxonomy: the first dimension is based on the application domain, the second dimension is based on technical challenge (model design vs data vs task). They will provide contributions in systematically reviewing (1) Vanilla Transformers, Vision Transformers (ViT) and multimodal-specific Transformers; (2) multimodal tasks and applications (e.g. pre-training paradigms and downstream tasks); (3) the common design patterns across such models, and (4) open problems and research directions. Effectively, Xu et al. wish to outline the state of the art in transformer-based MML: they discuss the application of self-attention to the integration of language, vision, audio, and other modalities; the application of large-scale pretraining (such as ViLBERT or CLIP) that is the background of many contemporary systems.

Their survey contents bring out major developments. In the case of architecture, they come across the establishment of the original Transformer (Vaswani et al.) to images (e.g. ViT) and then to multimodal fusion (e.g. such models as VideoBERT).[1] As an example, they mention VideoBERT as the initial work on transferring Transformer to the multimodal tasks and state that it has led to many other pretraining models (ViLBERT, LXMERT, VisualBERT and so on). They put special emphasis on CLIP (Contrastive LanguageImage Pretraining) as a milestone: it casts an image classification problem as a text image retrieval problem, and zeroshot generalization is realized by pre-training with many modalities at large scale. Xu et al. remark that in the data domain, multimodal datasets have become too numerous and varied. Million-scale corpora (e.g. Conceptual Captions 12M, LION400M) have also accompanied traditional benchmarking e.g. (COCO, VQA, Visual Genome). They also mention new modalities: 360 videos, 3D motion capture, and multi-view platforms (referring to particular new datasets including Pano-AVQA and MultiBench of 10 modalities each). Regarding tasks, they span a range between those that are now standard (image captioning, VQA, speech recognition) to those that are of recent concern (vision-and-language navigation, financial data fusion, abstractive tasks such as metaphor or hateful meme detection).

One of the components of the findings by Xu et al. is the synthesis of design patterns. They describe the fact that Transformers are naturally flexible to multimodal inputs: using all of the inputs to each modality as tokens and encoding modality labels or positions, a single self-attention network can handle input of different modalities. They list advantages of transformers in MML: the multi-head attention provides sub-spaces of interactions between features, global self-attention captures non-local correlations between modalities, and the large capacity has ability to learn cross-modal knowledge. The architecture decisions (single-stream vs two-stream networks,

pipelined vs joint) are also compared and it is pointed out that there is no one design that does all things well.

Importantly, Xu et al. end up with open challenges. They underline the idea that existing multimodal Transformers tend to become specialized to one modality or task and whether it is possible to obtain a universal model, which can handle any mixture of modalities and tasks, is an open problem. They point at issues such as semantic alignment (e.g., aligning recognized objects within an image with words in text) as being, in a sense, a still unsolved problem above auxiliary detectors or computationally expensive approaches. They observe the scaling problems (sequences of longer inputs increase the quadratic cost of the transformer) and data problems (data in the real world are noisy and weakly aligned multimodal). As an example, they state that vision-language pretrained models may perform poorly on fine-grained tasks such as detailed VQA or instance-level classification. These constraints highlight the regions of further research.

Regarding relevance, Xu et al. give specific knowledge that is viable in the agricultural structure. They are quite clear conceptually to extend to our field: e.g., an autonomy navigating robot would require multimodal sensors . . . e.g. camera, LiDAR, radar, [GNSS], HD map". Likewise, satellite imagery (camera), weather radar (radar), GPS/remote sensing (GNSS), and maps have to be employed in our system. The focus of the review supports the use of vision-and-language foundation models (such as CLIP), which implies that we would be able to use the textual labels or geotags in the pretraining of remote images as well. They also refer to video and geometrical data (360 2 video QA) suggesting the existence of models dealing with spherical or chronological series of images that are normally useful in time-series of satellite imagery. Transformer architectures based on the survey that they provide (ViT, VideoBERT, etc.) guide our architectural decisions: a Vision Transformer would encode spatiotemporal patches of satellite data, with a cross-modal attention layer combining in climate sensor tokens. These benefits of transformers that are recorded (e.g. global focus, scalability in terms of tokenization) portray that this practice of using a transformer backbone to integrate geospatial images with a time series is promising.

However, there are also flaws in the survey conducted by Xu et al. in comparison with our objectives. It is general and not specific to any particular domain so it does not concern things like encoding continuous geospatial coordinates, or managing missing sensor data. They mention these as open problems e.g. treating point clouds or continuous inputs in the transformer is not trivial which means we will have to make custom embeddings on climate/time data. The survey also presupposes that large corpora are at hand, which is not the case in agriculture where labeled multimodal data is hard to come by. Xu et al. cite the existence of so-called weakly-aligned data (e.g. e-commerce images with poorly-defined text tags) as a problem, and this bears uncanny resemblance to the fact that our satellite images, weather records, and socio-economic surveys might not fit together so well. Lastly, they are specific to transformer architectures and, as such, they fail to specify task specific approaches to structured prediction or multi-horizon planning that we will require to support decision-making. Those gaps mean that although we have the option of using the models that have been developed by Xu et al. (e.g. large pretrained multimodal transformers), we still need to add the domain knowledge and data to the work in regard to spatiotemporal consistency and interpretability.

The two surveys explore the multimodal AI but their thematic focus and methodology differ. Conversational agents are the focus of Sundar and Heck (2022), who analyse the dialogues at the turn, coreference, and multi-sensory levels. Their article systematically reviews task decomposition (representation, alignment, mix-and-match strategies), provides the state of the art in dataset subreddits and gives an account of the available corpora. The survey by Xu et al. (2023) travels over a much larger landscape of multimodal transformer-based models, avoiding domain specificity most of the time. They place the main focus on grouping the results according to the model architecture and training paradigm as transformers are considered a unified framework. Sundar et al. are application-centric in their viewpoint of the conversational AI applications and lay more emphasis on language-vision fusion, whereas in a more architectural lens, Xu et al. prefer to emphasize on the large-scale pretraining and model design.

Despite the above-mentioned differences, the two surveys have a number of themes in common. The critical role of representation learning is argued by both: the vision and language are jointly embedded (Sundar and Heck) and mixes of tokens are encoded naturally in transformers (Xu et al.). Both concentrate on the issues of fusion and alignment; one defines alignment as finding modal correspondences whereas the other mentions semantic alignment (fine-grained) is vital to vision-language tasks. Both accept the inadequacy of the current practices: On the one hand, Sundar explains that available methods are limited to maximizing individual QA assignments, not entire discussions, and on the other hand, Xu et al. state that most of the models are task-specific, and they are yet to find a universal multimodal learner. They, too, emphasize the expansion of data, Xu et al. list extensive multimodal data catalogs, and Sundar and Heck catalog video-text conversational corpora.

There are other differences which are seen through architectural decisions. Sundar and Heck list various network topologies, RNNs, CNNs, early multimodal autoencoders, and give Transformers comparatively little weight.[3] Their examples contain deep Boltzmann machines, LSTMs, and retrieval systems like TransResNet. In comparison, Xu et al. concentrate on Transformer-based models specifically and ViT, VideoBERT, and BERT-style cross-modal encoders in particular, and they follow the surge of their emergence in multimodal realms. This imbalance notwithstanding, both papers acknowledge the rise of Transformers; Sundar et al. state that the success of Transformers in text modelling has led to an increased number of vision-language application paradigms based on the notion of transformers. Therefore, even though Sundar & Heck question more classical and application-specific models, Xu et al. provides a modern and combined picture.

In a multimodal-AI perspective, Xu et al. have a wider scope of application including in agriculture. Their survey targets architecture general-purpose models and pre-trained models of large scales which can be easily transferred cross modalities. The work of Sundar & Heck is more confined to the level of conversational interactions having no direct connection with land management. However, their experiences related to the aspects of co-learning and the design of datasets have practical lessons.[24] Their plea regarding consistency in modalities to be reasonably interpreted can be extended to the realms of non-verbal discourse, and their case of inter-modal pooling of resources is the forerunner of mixing and matching non-monolithic agricultural data.

Judgments concerning its applicability in agricultural decision-making, bring out

areas of divergence. The paths that Xu et al. provide are quite practical: the examples of multi-sensor data to navigate use as a reference the spatial Temporal inputs typical of smart farming, namely satellite imagery and LiDAR associated with GNSS and sensor time series.[5] The video-language and vision-language models that they survey propose similar techniques to the satellite imagery sequences in addition to textual reports (e.g., descriptions of a crop). In their works on zero-shot and transfer learning (e.g. CLIP), they show promising potential that machines do not need retraining to transfer to new areas or classes of crops.

The two pieces balance out when it comes to limitations. Sundar & Heck identify data availability and task constraints, the unavailability of multi-turn, natural, multimodal dialogues, seeing large amounts of training data and the unfavorable lack of overall training aims. Xu et al. identify weaknesses in the designs and methodologies of models: the lack of universal architectures, alignment challenging nature, and efficiency. Sundar et al. do not pay much attention to the particular strategies to be adopted to modern architectures, but Xu et al. do not focus on outlining sequential decision cycles or challenges that discussions can have. Overall, a similar analysis of these questionnaires highlights the issue of co-learning and reflection underlying the major challenges of multimodal AI. Both of the surveys have important insights; in combination, they create a powerful picture of the field. Sundar et al. place multimodal AI in the context of conversations and prioritize the real integration of modalities, decomposition towards tasks, and sharing of resources by the modalities. Xu et al. place multimodal AI in the context of transformerbased architectures with the emphasis on large-scale pretraining, methodological progress and transferability. It will be in the future research and applications useful to interweave these strands.[25]

El Sakka, M., Ivanovici, M., Chaari, L., and Mother, J. (2025) present a full literature review on the use of convolutional neural networks (CNN) in smart agriculture. The contribution titled as, a review of CNN applications in smart agriculture using multimodal data, combines the results of over 115 recent papers to discuss the applications in identifying weeds, identifying diseases, crop classification, water management and prediction. One of the main contributions is proving the way CNNs could be used to take advantage of the multimodal input, such as RGB, multispectral/thermal and radar imagery in the context of an Agriculture 5.0. The following paper takes a critical look at the most recent writings related to the use of convolutional neural networks (CNNs) in the agricultural setting. The authors consider that systematic reviews do not consider the data sources heterogeneity and, respectively, their methodological variations. To be more precise, the previous syntheses mostly do not take into account the presence of multimodal data input or explain the used architecture of CNNs in detail. To counter it, the given paper conducts a systematic review of articles of the 2018-2024 co published in the context of the explicit matching of various data modalities with particular acquisition methods. This limited research focus is supposed to clarify the level to which data modality as well as the methods choice have an impact on the emergence of farm AI solutions

In the current undertaking, a two-staged approach to literature review will be used. Phase 1 implies a bibliometric study of the Web of Science by using the key query of convolutional neural networks and agriculture, showing about 27,700 documents. Following the filtering of conference/journal papers in the English language, 733 core

papers are left. These news articles are subjected to mechanical trend studies (e.g. word development). Simultaneously with it, in the Phase 2, a manual literature survey is conducted. Some of the pertinent articles have been chosen in Google Scholar, IEEE Xplore and other websites by giving priority to the articles that meet both the requirements of relevance and diversity. A categorization of works is then done upon identifying five principal areas of smart agriculture namely weed detection, disease detection, crop classification, water management and yield prediction. The paper has a table 1, which sums up the articles based on categories. The paper continues the review of modality of input data (RGB, near-infrared, multispectral, SAR, thermal, vegetation indices) and backgrounds of frequent CNN architectures and evaluation measures.[24] Such a systemic approach will allow processing an investigation of the application of various CNN models, namely, AlexNet, ResNet, YOLO, and others, in such domains. The review’s findings show that CNN-based methods generally achieve high accuracy and outperform traditional techniques across agricultural tasks. Key results include: In traditional CNNs, the ability to differentiate between the weeds and crops does not vary significantly in regard to the model structure; several publications reveal that some such models as YOLO and Faster R-CNN are functionally similar. Once, UAV-based image data were annotated and employed alongside CNN, which returned the value of classification accuracy a success of 92 percent, which is higher than was achieved by a Random Forest classifier being trained with the same data.[24] The accuracy rate of classification reached above 99 percent in a parallel experiment in comparisons between CNNs and SVM/RF/AdaBoost classifiers. A further analysis included extra spectral bands and the Normalized Difference Vegetation Index (NDVI), and thus largely increased the general performance of the CNN-based system.[7]

Robust performance VGG, ResNet, Inception are deep convolutional neural network (CNN) classifiers that have shown good results when used on leaf images with an overall accuracy of over 95%. Many studies mention the precision and recall rates of around 9899 % in the classification of healthy and diseased samples of the plants. Similarly, Mask R-CNN and UNet segmentation networks have been integrated with CNN blocks to locate disease regions with a performance of close-to-perfect accuracy (and it was seen in UNet+CNN that it reached the accuracy of 99 %). By incorporating non-visible wavelengths in multispectral imaging, one would be able to augment the spectrum available to detect disease: the single multispectralCNN model was able to detect the presence of blight in the tomatoes 36 h before the visible symptoms constituted its occurrence, and the accuracy was around 91 %. Moreover, a combination of RGB and a near-infrared imaging system exhibited nearly 93 % accuracy. Taken together, the results prove the potential of the deep learning methods to classify healthy and diseased leaves with a great amount of fidelity and the additional usefulness of multispectral data streams.

Traditional convolutional neural networks (CNNs) have proven to be better at classification when applicable in various crop-identification assignments. In particular, a hybrid model CNN-recurrent neural network reached the level of accuracy of 95 percent when classifying wheat varieties, based on leaf images, and CNNs trained via transfer learning performed at around a 99 percent accuracy on a variety of crop leaves. Finally, when CNN RNN architectures were deployed in multispectral time-series Sentinel-2 data, they performed better than traditional classical classifiers (support vector machines and random forests) and achieved an overall accuracy

of 93 per cent (IoU 79 per cent of the tea plantation), on large-scale land-cover mapping. The combined application of both multispectral imagery and a batch of synthetic-aperture radar data was also of great benefit in said studies based on their spectral coverage and cloud-penetration attributes respectively. Convolution Neural Networks (CNNs) have however, become a favorite technique in determining the water stress of plants and determining the soil moisture. Being the light cleaners, the lightweight CNNs have proven to classify images resolving drought levels with an accuracy of around 87 percent in image-classification exercises. Moreover, CNN LSTM hybrids that integrate spatial and temporal aspects have attained a high rate of more than 98 percent in predicting irrigation drugs on chickpea plants. Very early stress can be detected through hyperspectral input; multiple 1D-CNN produced 96 per cent accuracy with pre-symptomatic drought stress on tomato seedling spectra. In general, it was reported that a combination of color pictures with spectral indices (NDVI, thermal) led to the improvement of the water-stress detection ability. It has been revealed that CNN-based regression models have shown a significant increase in crop-yield predictions that involve UAV or satellite imagery or other supplemental data. In one empirical study a CNN using weekly weather data (precipitation, temperature, etc.) in conjunction with UAV multispectral images performed better than linear-regression baselines. Another study based on the same idea stated that a CNN based on the multispectral imagery was much more effective than traditional regressors, such as linear, random forest, and gradient-boosted-tree regressors, in predicting the wheat yield. In parallel, spatio-temporal CNN-LSTM models (similar to hybrid ones except that the temporal blocks are LSTMs instead of CNN) demonstrated high performance on long multi-year remote sensing time series.[21] In all those domains, the synergistic advantages of multimodal data can be seen. As an example, the process of the separation of weeds worked better when the RGB imagery was combined with NDVI/NIR channels and the process of the drought detection worked better when it had thermal or spectral channels. Most reliable predictions in yield modeling were observed upon blending of imagery and weather and soil parameters. A comparative evaluation has revealed that the CNN are always superior to the classical ML approaches such as support-vector machines (SVMs) and random forests where there are lots of images involved. On the other hand, it was noticed that in case the tasks are tabular or engineered, tree-based techniques (e.g., XGBoost or RF), tend to perform better overall on average. All in all, the authors accept the conclusion that 2D CNNs, 3D CNNs, and CNN-LSTM hybrids are suitable in the setting of complex spatio-temporal data, which makes them especially appealing in agriculture.[31] The current report provides a logical evaluation of the scientific article Machine Learning in Agriculture: A Review (2018, 18, 2674) in Sensors. The article written by Konstantinos G. Liakos et al. provides an all-presenting exposure of the key applications and the related issues of machine languages (ML) in the sphere of modern agriculture. The review methodically questions the growing importance of machine learning in the agricultural field and concludes this traction to the recent advances in big data approaches and extreme computing. The article describes and analyses various ML applications in different agri-technological fields, therefore, providing a unified description of the sphere until 2018. The reviewed articles have been, on a first level, classified in four generic categories: This section delves into the use of ML for critical tasks such as yield prediction, early detection of crop diseases, identification and mapping of weeds, optimization

of irrigation schedules, and precise nutrient management for enhanced crop health and productivity. The paper highlights ML’s utility in monitoring livestock health, predicting and detecting diseases, analyzing animal behavior, and optimizing production parameters like feeding and breeding.

Applications related to soil are discussed, focusing on predicting various soil properties (e.g., moisture, organic carbon, nitrogen content) and guiding fertility management strategies. The review addresses how ML can be leveraged for intelligent irrigation scheduling and real-time monitoring of water quality in agricultural settings. The authors touch upon the integration of ML with autonomous agricultural vehicles and robotic systems for tasks such as precision spraying and harvesting, paving the way for more efficient and labor-saving farming practices. The paper also provides an overview of commonly employed machine learning algorithms, broadly classifying them into supervised learning (e.g., Support Vector Machines, Artificial Neural Networks, Decision Trees, Random Forests for regression and classification tasks), unsupervised learning (e.g., K-means for clustering), and emerging deep learning techniques. Furthermore, it addresses the challenges inherent in implementing ML in agriculture, such as data scarcity, data quality issues, the need for domain-specific expertise, and computational demands. The authors conclude by outlining future research directions, emphasizing the importance of developing more robust and interpretable ML models, integrating diverse data sources, and fostering interdisciplinary collaboration.[1]

This study is a systematic overview of the application of machine-learning (ML) technology in the agricultural sector—a rich content reservoir for academia and practice alike. The categorization of various initiatives by well-organized thematic blocks enables the review to have an organized understanding of a fast-developing field. Its key contributions can be divided into three: organization of the heterogeneity of the available applications; description of the methodological diversity of agricultural ML, identification of major practical and theoretical challenges. In addition, the debate identifies future directions of research in which more focus should be given scientifically.[4] The breadth of coverage is that the review covers the full range of agricultural subdomains namely crop and soil management, livestock production, precision farming, automation and others, giving a comprehensive overview. The story has a logical structure that can be navigated both in contents and in ideas. The relevance of ML in agriculture is quite topical since the world is grappling with food security and sustainable production that is increasingly becoming difficult to meet. The broad bibliography demonstrates the high requirements of the research foundation and the continuation of scholarly research.

Possible Synthesis Improvements: In spite of these strengths, there are a number of areas which could use some expansion. To begin with, more detail may be added to the description of the personal algorithms and in particular, their relative advantages / disadvantages, and their relevance to particular problems in agriculture. Second, new paradigms, like Reinforcement Learning in agricultural robotics, Federated Learning to deal with decentralized data and powerful Generative AI models, are not covered, even though their break-out could have been predicted. Third, the review would be enriched in terms of the economics, its economic viability, cost benefits and socio-economic impacts and the impacts on labour, and digital divide. Lastly, the data issues segment: standardization, interoperability problems, and possible ways of solving the problems need improvement.[4]

Overall, the review gives a credible description of ML applications in agriculture, seamlessly combining a range of applications on the one hand and stating the important methodological, practical, and conceptual issues on the other. The epilogues drawn at the end provide a roadmap on future research attempts that is geared towards maximising the efficacy and influence of ML. In the review conducted by Liakos et al., some strategic pathways are still indicated in the development of the data-driven agriculture: Data ecosystems: The need to have high-quality, diverse and carefully curated datasets is still unquestioned. The main focus of research should then be building strengthened data-collection systems, building strong data-sharing systems, and the use of standardized data formats, all with the objective of making machinelearning models develop faster. Explainable AI (XAI): Machine-learning architectures are becoming increasingly complex (or opaque) and this tendency may hinder their adoption by industries that require transparent and interpretable outputs, e.g. agriculture. Research on XAI techniques applicable to agriculture is thus necessary in the future.

“Edge AI and Decentralized Computing: Real-world implementation of machine-learning that involves deployment to work on farms often requires edge computation as a bottleneck due to bandwidth limitations. Breakthroughs in low-power, efficient machine-learning algorithms, as well as designing suitable edge-computing architectures, are, in turn, important.[13] “Integration of IoT and Robotics:” Greater combination between machine learning and the Internet of Things (IoT) sensors to advanced robotics will allow more accurate, autonomous, real-time decision making in the agricultural setting. “Sustainability and Environmental Impact”: The future applications of machinelearning should increasingly focus on how to optimize the use of resources, i.e. water, fertilizers and pesticides, to reduce environmental impacts and curtail unsustainable farming activity. [15] Such a review introduces a comprehensive presentation of Machine Learning (ML) critical contributions to the enhancements of precision agriculture, which is widely referred to as smart farming. Based on the extensive survey by the team of Sharma et al. (2020), the discussion is developed in a systematic manner considering the key use cases, the major principles and the existing constraints of the ML implementation into modern agriculture. With the growing rate of population in our world, frequent changes in weather patterns and a decline in non-renewable natural resources, integration of precision agriculture and the Internet of Things (IoT) appears as the promising diet on the way to high sustainability and productivity simultaneously. Each review paper shows a variety of substantive ML-based applications produced by them, such as the prediction of crop yields, disease diagnosis, weed control, and soil quality analysis, and discusses them in terms of their ability to streamline the use of resources and lessen the degrees of environmental disturbance. It also examines ongoing data management and model complexity and practice implementation issues and formulates the vision of what future research directions will be to perfect the use of ML and its application to agriculture. Agriculture is the pillar towards economic growth and food security globally. Nonetheless, modern agriculture faces new and challenging issues never encountered before, with one of them being the task of feeding the growing global population, addressing the new and mostly unpredictable climate conditions, and the limited availability of resources, including arable land and fresh water. The conventional practice of farming that entails a uniform concentration of inputs has often led to wastefulness, degradation of the environment and low

production. In this case, it is necessary to use new tools and strategies that ensure sustainable agricultural output. Precision agriculture, or to be more general smart farming, can be discussed as a paradigmatic answer to the problem as it is a data-driven, site-specific method of managing the resources and the way operations are to be carried out.[15]

The success of precision farming depends on the advanced processing and analysis of huge amounts of information generated by the various non-homogenous platforms. Machine Learning (ML) is one of the key enablers, in that context. In the terms described by Sharma et al. (2020), ML endows the machine with learning capabilities without explicit programming. This innate ability empowers the ML algorithms to recognize elaborate correlations, develop predictions and decisions based on a voluminous amount of agricultural data, between soil qualities and environmental variables to plant overall health imagery. In the synthesis with the Internet of Things (IoT) and those more advanced sensor technologies, ML is an impetus to bring the next farming revolution and that means producing higher yields, reducing wastes, and improving environmental care. The current study is a systematic review and synthesis of the major applications of machine learning in precision agriculture with heavy reliance on Sharma et al. (2020). The discussion is broken down into three parts: outlining the separate sphere of fluency in which the practical efficacy of ML has been demonstrated in precision agriculture; outlining the current ML techniques that are utilized in each of these spheres; and identifying the systemic issues that become apparent when implementing the ML use in agronomic contexts. Last but not least, it proposes a set of future research directions that can be used to develop the wider discipline.

Precision agriculture is a modern operational concept of farming management, which relies on spatiotemporal variations, which require systematic observation, measurement, and responsive adjustment in and between crop stands. The most explicit and important aspect of this method is the rule of right place, right time, right amount, which is aimed at abandoning traditional similar models of management in favor of a more heterogeneous one by going out of their way and making use of the natural diversity of the agricultural landscape. The main effects that are aimed at by PA are the concurrent increase of the production of crops, the decrease in the impact on the environment, and lower costs of inputs (Sharma et al., 2020).[20]

The successful implementation of precision agriculture is also predetermined by the simultaneous implementation of a set of modern technologies aimed at the simplification of the process of obtaining, processing, and implementing data.

2.3 Summary of Key Findings

Multimodal machine learning is a set of techniques aimed at receiving, mapping and fusing multiple heterogeneous data streams to obtain a comprehensive assessment of complicated systems. Specifically, within this agricultural construct the generic representatives are classified into three unique environmental dimensions;

Geospatial and Reconnaissance visual features Ground-level field photographs as well as segmented environmental frames can be the main input of deep Convolutional Neural Networks (CNNs) to predict local surface properties and classified taxonomic information at a fine-scales over large areas in disparate landscapes.

Categorization of Climatic parameters and environmental variables: Multi-temporal

time-series records (i.e., rainfall parameters, temperature dynamics, as well as solar radiation indicators) are estimated through tree-based ensemble architectures when linked with retrospective agricultural functions to facilitate extrapolation over a temporal range in conjunction with historical anthropogenic influences.

Physiographic and Administrative Variables: High dimensional tabular datasets containing aggregated gridded soil properties (e.g. mean regional pH, clay fractions, nitrogen concentrations) combined with categorical macro-regional administrative district identifiers that govern geographic crop zoning

Contemporary multimodal architectures interact with various datasets by organizing fusion topologies into different paradigms where the points of intersection between streams of information are:

Early Fusion (Feature-Level): The concatenation or mathematical combination of the original or intermediate feature vectors extracted from independent modalities which can then be fed into a single joint learning network.

Late Fusion (Decision-Level): a cautious modality fusion setting with domain-specific models separately processing their respective modalities in full isolation. These distinct output vectors (like independent probabilities for computer vision classes and tabular regression predictions) are then aggregated via rules, voting heuristics or scripted policies to produce a single final recommendation.

Hybrid Fusion: A two-phase method that combines both low-level feature fusions (early fusion) and high-level structural ensemble methods (late fusion) in the same model pipeline.

Depending on their formulation, these modalities allow machine learning frameworks to overcome the restrictions of single-modal judgment methods via cross-modal logic automation. For research datasets that are constrained or unpaired, where direct end-to-end alignment is not an option, modular decision-level late fusion serves as an empirically justified guardrail. Thus, ensuring safer highly interpretable decision logic shielding end-users from risky farming recommendations and preserving structure stability against representation collapse.

Table 2.1: Thematic position of the present thesis within related agricultural machine learning work.

Theme	Data	Typical method	Gap / relevance
Soil or crop image classification	Field imagery	CNN	Strong visual classification; limited direct yield context
Yield forecasting	Weather, satellite, soil, historical yield	Regression, RF, XGBoost	Predictive but may ignore visual soil evidence
Decision support	Model outputs + rules	Rule-based	Interpretable but needs validated priors
This thesis	Soil image + district tabular yield + suitability rules	CNN + XGBoost + decision-level fusion	Prototype for Bangladesh Aman/wheat recommendation

The literature shows that soil image classification, yield forecasting and agricultural recommendation have been investigated independently. This paper addresses the research gap of aggregating these components into one transparent Bangladesh–

specific workflow. The thesis does not set out to solve all problems in precision agriculture; rather it shows how data at hand can be integrated into a modular recommendation prototype where the performance of its components are computable and their limitations being clearly declared.

Chapter 3

Methodology and Project Management

This chapter describes the architectural specifications, functional workflows and impact matrix for the developed data driven agricultural decision-support system. It is designed around the Bangladesh agrarian context, with no living chemical sensors or GPS tracking that violates privacy. The isotopically-informed manual district selection effectively provides this geographic identity, allowing for two independent analytical streams (an EfficientNet-B0 CNN for visual soil taxonomy and an XGBoost ensemble for spatiotemporal yield forecasting) joined through a transparent 0.55 confidence threshold and an auditable rule engine (`suitability_rules.json`).

3.1 Final Specifications and Requirements

This paper presents a data-driven agricultural decision-support system for crop-based farming with a focus on low hardware-dependency. The system outputs fast and with high accuracy real-time predictions for Aman and Wheat crops, using an intuitive Streamlit dashboard integrates fast and accurate real-time predictions for Aman and Wheat crops in a hybrid ML framework, combining a CNN-based EfficientNet-B0 (pytorch interface) soil image classification model with tabular yield forecasting based on an XGBoost framework [22], [24]. Trained only on 1,479 original soil images [38] (with CyAUG augmentation at training only) and historic agricultural tabular data [33], [35], [37], without live chemical sensors and raw GPS hardware.

1. **System Requirements and Tech Stack:** The core machine learning architecture includes PyTorch for CNN soil classification and sklearn/XGBoost for tabular crop yield forecasting. Users interact through a Streamlit web dashboard (`gui/app.py`) for visualization and real-time predictions. The application is optimized for GPU training and standard CPU inference.
2. **Functional Specifications (Core Modules):** An EfficientNet-B0 soil classification engine assigns one of seven taxonomy classes (plus flooded land) with a conservative 0.55 abstention threshold. The yield forecasting engine predicts Aman and Wheat output using district, season/year (2026 planning horizon), climate features, and integrated flood/environmental metrics. A decision-level

fusion system reads `suitability_rules.json` and outputs suitable, conditional, or not_recommended guidance.

3. **Data Requirements:** The Soil-Classification-Dataset provides 1,479 original images; CyAUG synthetic images are used only during training. The tabular pipeline uses `Final_Integrated_Dataset_with_Soil.csv` with historical yield, district statistics, and gridded soil priors (pH, clay). No live chemical sensors or raw GPS data are required; users select district manually.

3.2 Societal Impact

- **Improving food security:** District- and season-level yield forecasts help farmers plan stable food supply and prepare for shortage risk.
- **Disaster mitigation and resilience:** Flood-aware inputs help farmers anticipate climate-related agricultural losses.
- **Empowering farmers with data:** The Streamlit GUI turns climate, satellite, and soil data into actionable crop and management guidance.
- **Automated soil classification:** CNN-based soil taxonomy supports optimal crop selection and sustainable practice.
- **Data-driven policy making:** Aggregated modeling can inform government and agricultural institutions and support rural economies.

3.3 Environmental Impact

- **Promoting sustainable farming:** Avoiding over-cultivation of unsuitable land reduces soil degradation.
- **Resource optimization:** Accurate yield predictions support better fertilizer and water management.
- **Climate change adaptation:** Climate and flood data support adaptation to emerging environmental patterns.
- **Lower crop loss:** Better crop choice reduces wasted energy and emissions from failed production cycles.
- **Ecosystem preservation:** Data-supported land management reduces pressure to clear new land for crops.

3.4 Ethical Issues

- **Digital divide and accessibility:** The Streamlit GUI requires a smartphone camera and likely internet access, which may widen the technology gap between resource-rich and rural farmers.

- **Data bias and fairness:** Training data biased toward certain districts may reduce accuracy for less-represented regions.
- **Algorithm Transparency and Trust:** No clear model limitations and responsible communication when recommendations are ineffective for complex CNNs and XGBoost models.
- **Economic Market-Level Effect:** A single universal high-yield recommendation is likely to give the recommendations widespread adoption, ultimately impacting market prices and crop diversity.
- **Data Privacy and Sovereignty:** The specific user location where they farm, their soil, and crop choices raise huge data privacy and potential abuse concerns.

3.5 System Workflow Diagram

To summarize, the end-to-end system workflow from user input via dual predictive models to rule based fusion is depicted below [22], [29].

Figure 3.1 shows how GUI inputs flow through the CNN and XGBoost into the fusion layer.

3.6 Project Management Plan

The project flow in four phases.

Stage 1: User Inputs (Streamlit GUI Frontend)

The user-facing dashboard obtains three data inputs, without external tracking devices: district selection (geographic identity), season (planning 2026 cycle), and one soil picture from CNN taxonomy analysis.

Stage 2: Data Processing and Two Predictive Models

Inputs separated into parallel streams: a PyTorch EfficientNet-B0 soil classifier (only trained on the training split with CyAUG) and an XGBoost yield engine that combines historical yields, climate features, flood history, and gridded soil priors for Aman and Wheat.

Stage 3: Transparent Decision-Level Fusion using Rules

CNN and yield outputs enters into a dual-path fusing layer using 0.55 confidence. The system will abstain below threshold, above threshold a rule engine will read `suitability_rules.json` and integrate soil class, yield forecasts and suitability tiers. Formal fusion equations are given in Section 4.2.12.

Stage 4: Final Output Generation

Paths of high confidence display classification, suitability tiers, and expected yield, while paths of low confidence post an abstention notice that requests a clearer picture of the soil.

3.7 Risk Management

- **Data integrity and validation risks:** No paired image–yield ground truth and no field validation with farmers.

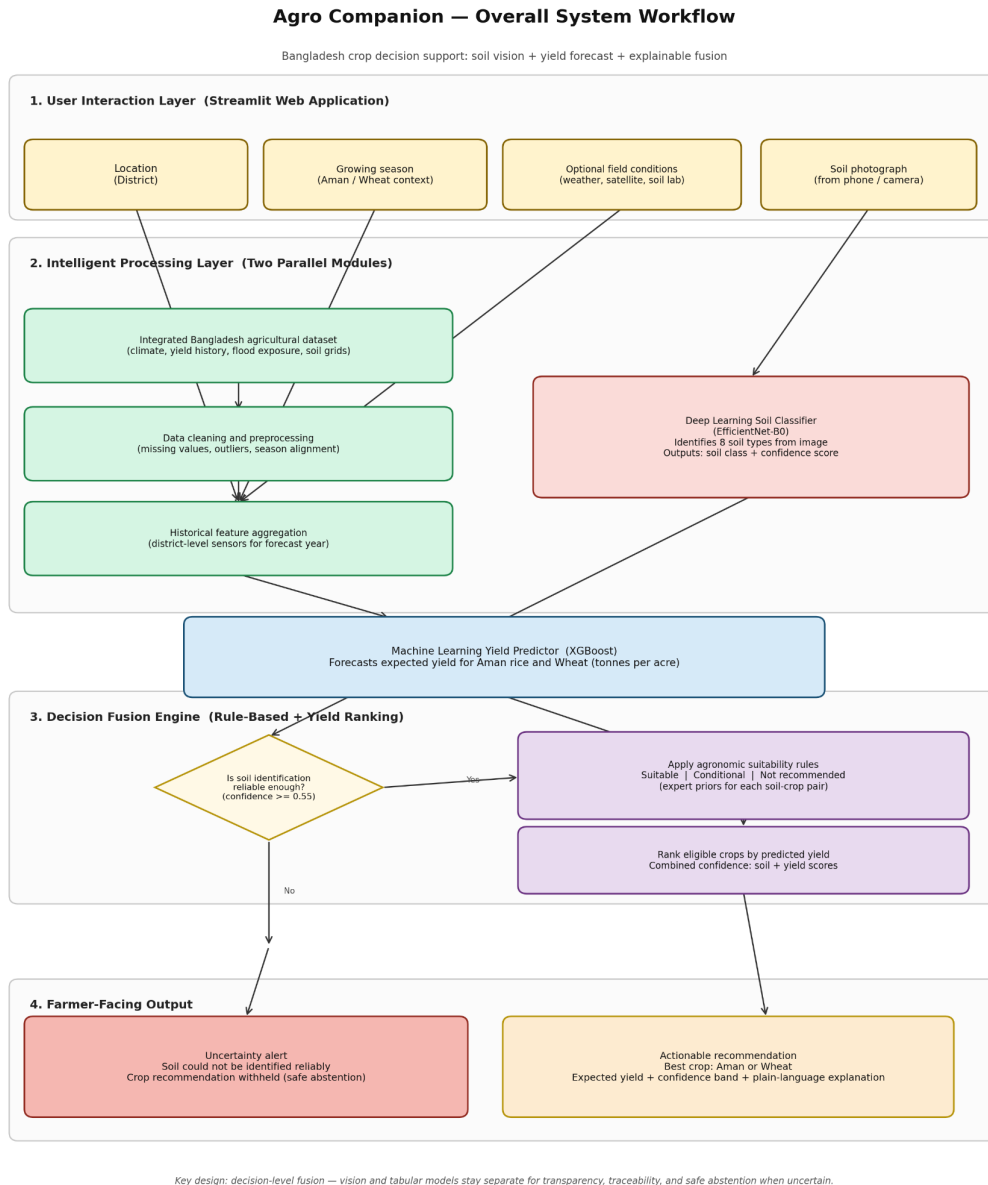


Figure 3.1: System workflow diagram for the Agro Companion decision-support pipeline. Four stages: input, dual models, fusion, and output.

- **Management:** Index-aligned proxy pairing is disclosed; the tool is labeled an undergraduate engineering prototype, not a commercial advisory product.
- **Model reliability risks:** Limited dataset (1,479 images), class imbalance, and weaker wheat performance ($R^2 \approx 0.51$ vs. 0.85 for Aman).
- **Data augmentation:** CyAUG is applied during training only, never on validation or test splits.
- **Confidence thresholds:** A 0.55 soil-confidence threshold triggers abstention in roughly 8% of paired cases.
- **Honest evaluation:** Strict 2025 temporal holdouts avoid optimistic reporting.
- **Crop–soil suitability rules:** Literature-based rules in `suitability_rules.json` are auditable without retraining CNNs.
- **Usability risks:** Soil chemistry is not inferred from photos; correct district selection is required via mandatory GUI fields.

3.8 Economic Analysis

- **No direct profit claims:** Fusion performance is reported under a public holdout protocol; profit increases are not claimed.
- **Theoretical economic impact:** Mitigating crop failure risk can reduce inappropriate investment in seed, fertilizer, and labour.
- **Low-cost deployment:** The prototype uses free historical data, satellite/climate priors, and smartphone soil photos without chemical sensors.
- **Prototype status:** Field validation and multi-season ROI analysis remain future work.

Chapter 4

Design Process and Model Specification

Design and Structural Implementation of Multimodal Machine Learning Platform architecture where the second layer (the prediction layer) is optimized for predicting accuracy across many different environmental streams knowing only the expensive field sensors or active GPS monitoring. The aim of this chapter is to introduce the algorithmic and spatial methods that will take raw Geospatial, Aman rice production and operational decision-support pipeline from climatic, image, and tabular data to wheat production in Bangladesh.

4.1 Design Process

The paper introduces multi-modal machine learning platform to consider sustainable agricultural land management in Bangladesh. The overall workflow follows:

data acquisition → feature engineering → multimodal integration
→ modeling → evaluation and interpretation

We choose the administrative district boundary (typical spatial unit) to enable the temporally heterogeneous datasets comparability. The following indicators at district and year/season level are extracted directly from satellite based features (NDVI and SAR features) derived from Sentinel missions. For climate data, point data at district centroids is extracted from the NASA POWER API [37] and aggregated to typical agronomical seasonal windows. Soil and structured properties from SoilGrids [35] are combined into an integrated 2018–2025 dataset. Yield records are sourced from national agricultural statistics [33].

A parallel CNN pipeline classifies visual soil texture from field photographs using transfer learning. The image modality outputs categorical soil labels that feed the decision-level fusion layer together with geospatial, climatic, and tabular features.

4.1.1 Feature Correlation Patterns

The heat map of correlation analysis reveals relationships among key variates:

- NDVI usually has a positive, although not strictly linear, relationship with rainfall.

- Extreme temperatures may relate negatively to vegetation response depending on season.
- Climate features are not redundant—they explain additional variation in vegetation and yield signals.

These relationships justify multimodal feature selection that combines rainfall, temperature, and NDVI rather than remote sensing alone [14], [15].

Figure 4.1 summarizes pairwise Pearson correlations among integrated tabular predictors used before yield model training.

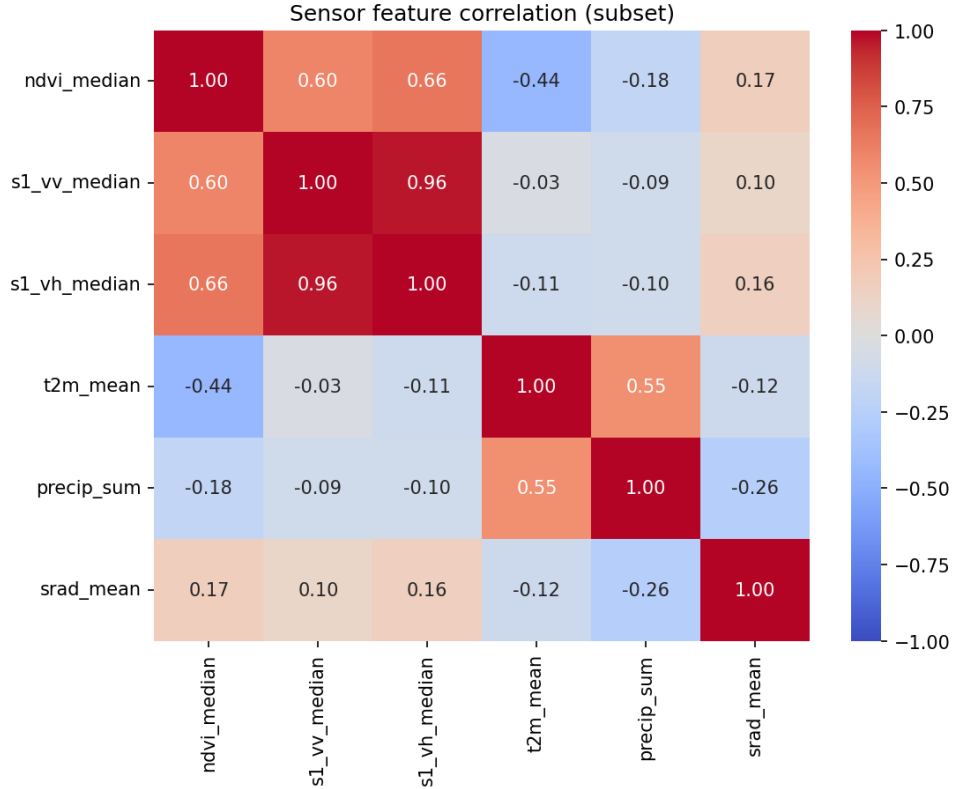


Figure 4.1: Feature correlation heatmap for integrated predictors. Stronger absolute correlations guide feature selection and redundancy checks.

4.1.2 Temporal Trends (NDVI and Temperature)

Time-series inspection reveals discernible seasonality in NDVI and temperature over the seven-year analysis window [34]. These temporal behaviors justify season-aggregated climate predictors, inclusion of year/trend terms, and multi-year evaluation rather than single-season fitting.

Figure 4.2 shows how many district-year records are available per season, supporting the temporal holdout design.

4.1.3 Spatial Performance Differences

The EDA identifies the districts with the highest performance metrics according to vegetation and yield signals, as evidenced by spatial units of productive potential [33]. That helps in Adaptive Decision Making and district-level baselines.

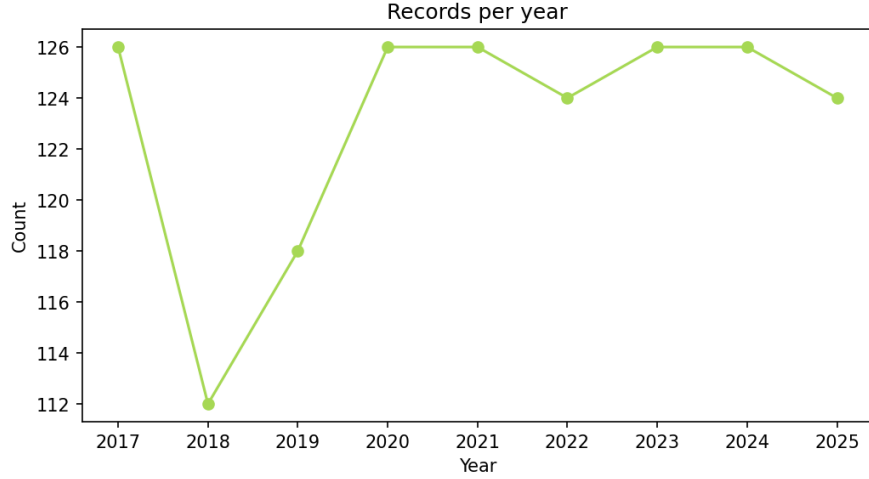


Figure 4.2: Temporal record distribution and seasonal data availability by year. Confirms sufficient multi-year coverage for train/test separation.

Figure 4.3 highlights districts with consistently higher or lower yield across the study period.

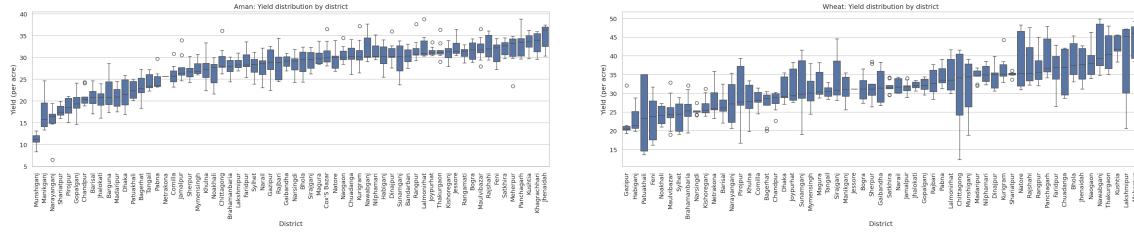


Figure 4.3: Spatial performance differences across districts. District-level yield distributions for Aman (left) and Wheat (right).

4.1.4 Data Integration Outcome

This integrated dataset consists of satellite properties (Normalized Differential Vegetation Indicator (NDVI), SAR (Synthetic Aperture Radar) metrics, climate metrics (Temperature, Precipitation) (2018–2025) as well as district level soil characteristics. This enables both exploratory analysis and predictive modeling in different climates.

4.1.5 Results of Data Processing and Exploratory Data Analysis

Data cleaning and valid-target enforcement: Non-positive yield records are immediately removed as such records are invalid targets.

Missing-data handling (cloud cover): Missing NDVI values are imputed with IterativeImputer using radar and climate relationships.

Feature scaling: StandardScaler normalizes predictors before training.

Outlier diagnostics (Aman yield): Extreme values of yield are excluded up to a plausible agronomic range to deem flood zeros and Misreport.

4.2 Preliminary Design and Model Specification

4.2.1 Notation

Table 4.1 defines symbols used in the equations below.

Table 4.1: Mathematical notation for model specification and fusion.

Symbol	Meaning
c_{soil}	Maximum softmax probability from the soil CNN
C_{yield}	Yield-model confidence heuristic
C_{final}	Fused recommendation confidence
$\hat{y}_c$	Predicted yield (tonnes/acre) for crop c
$R(s, c)$	Suitability category from <code>suitability_rules.json</code>
$\mathcal{E}(s), \mathcal{X}(s)$	Eligible and excluded crop sets for soil class s

4.2.2 System Inputs (Modalities)

1. **Geospatial/remote sensing features:** NDVI and SAR-derived vegetation and surface indices.
2. **Climatic features:** Seasonal temperature, precipitation, and solar radiation summaries.
3. **Soil and ancillary structured features:** Gridded soil properties and district-level predictors.

4.2.3 Target Variables (Outputs)

Primary targets are district-level yields for Aman rice and wheat. Although area and production are available, yield is the main supervised target as the most direct productivity indicator.

4.2.4 Agronomic Season Rule (Crop-Specific Filtering)

Aman rice uses the monsoon window (June–September). Wheat uses the non-monsoon window (October–May spanning two calendar years). Climate features are extracted accordingly.

4.2.5 Baseline Modeling Assumption (Preliminary)

Supervised regression maps multimodal predictors to crop yield. Linear regression provides a transparent benchmark before nonlinear models. The *deployed* yield models use a temporal holdout split (holdout year 2025), not a random 80/20 partition used only in baseline comparison notebooks.

For district d , season s , and target forecast year T , historical rows are

$$\mathcal{H}_{d,s,T} = \{\text{rows} \mid \text{district} = d, \text{season} = s, \text{year} < T\}. \quad (4.1)$$

Sensor and climate features are aggregated by the within-history median:

$$\hat{x}_j = \text{median}(\{x_{d,s,y}^{(j)} : y < T\}). \quad (4.2)$$

Meta-features for the forecast row are

$$\text{forecast_year_index} = T - y_{\min}, \quad (4.3)$$

$$\text{hist_n_years} = |\{y : y < T\}|, \quad (4.4)$$

$$\text{hist_yield_mean} = \text{median}(\mathcal{Y}_{d,s,T}^+), \quad (4.5)$$

where $\mathcal{Y}_{d,s,T}^+$ denotes positive yields in $\mathcal{H}_{d,s,T}$ (numbered (4.3)–(4.5)). With holdout year T_{hold} (e.g. 2025),

$$\text{Train} = \{\text{samples with target_year} < T_{\text{hold}}\},$$

$$\text{Test} = \{\text{samples with target_year} = T_{\text{hold}}\}.$$

Crop-specific prediction is

$$\begin{aligned} \hat{y}_{d,c} &= f_c(\mathbf{x}_{d,c,T}), \\ f_c &\in \{\text{XGBoost}_{\text{aman}}, \text{XGBoost}_{\text{wheat}}\}. \end{aligned} \quad (4.6)$$

4.2.6 Multimodal Feature Set (Final Feature Specification)

The tabular workflow uses NDVI median, S1 VV/VH medians, seasonal t2m mean, precip sum, srad mean, and SoilGrids pH, SOC, clay, and sand (0–30 cm).

4.2.7 Model Architecture

Three tabular algorithms are compared: linear regression (baseline), random forest, and XGBoost (deployed) [5]. The deployed XGBoost regressors minimize squared error during training:

$$\mathcal{L}_{\text{reg}} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2. \quad (4.7)$$

This is the objective minimized by `reg:squarederror` in the yield pipeline. RMSE (Eq. (4.13)) is the square root of this quantity; R^2 (Eq. (4.15)) is computed on the holdout set from residual and total sum of squares. Models are evaluated with the metrics in Section 4.2.8.

Soil Image Classifier (EfficientNet-B0)

For logits $\mathbf{z} = (z_1, \dots, z_K)$ over K soil classes [7], [24], softmax probabilities are

$$p_k = \frac{\exp(z_k)}{\sum_{j=1}^K \exp(z_j)}. \quad (4.8)$$

The predicted class and soil confidence are

$$\hat{y} = \arg \max_k p_k, \quad c_{\text{soil}} = \max_k p_k. \quad (4.9)$$

Abstention uses the deployed threshold 0.55:

$$\text{status} = \begin{cases} \text{uncertain}, & c_{\text{soil}} < 0.55, \\ \text{ok}, & \text{otherwise.} \end{cases} \quad (4.10)$$

With N_{train} training images and n_c images in class c ,

$$w_c = \frac{N_{\text{train}}}{K \cdot n_c}. \quad (4.11)$$

Cross-entropy with label smoothing $\varepsilon = 0.05$ uses smoothed one-hot targets $\tilde{y}_c = (1 - \varepsilon)\mathbf{1}[y = c] + \varepsilon/K$:

$$\mathcal{L} = - \sum_{c=1}^K w_c \tilde{y}_c \log p_c. \quad (4.12)$$

Input images are normalized with ImageNet statistics:

$$x'_c = \frac{x_c - \mu_c}{\sigma_c}, \quad \boldsymbol{\mu} = (0.485, 0.456, 0.406), \quad \boldsymbol{\sigma} = (0.229, 0.224, 0.225).$$

4.2.8 Evaluation Metrics

Training minimizes $\mathcal{L}_{\text{reg}}$ (Eq. (4.7)) for yield and $\mathcal{L}$ (Eq. (4.12)) for soil CNN. After training, holdout performance is reported with the following evaluation metrics (N test samples for regression, K classes for classification).

Regression (yield). Let $SS_{\text{res}} = \sum_i (y_i - \hat{y}_i)^2$ and $SS_{\text{tot}} = \sum_i (y_i - \bar{y})^2$. Then

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} = \sqrt{\mathcal{L}_{\text{reg}}}, \quad (4.13)$$

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|, \quad (4.14)$$

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}} = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}, \quad (4.15)$$

$$\text{Overfit gap} = R_{\text{train}}^2 - R_{\text{test}}^2. \quad (4.16)$$

Classification (soil CNN).

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[\hat{y}_i = y_i]. \quad (4.17)$$

Per class c ,

$$\begin{aligned} \text{Precision}_c &= \frac{TP_c}{TP_c + FP_c}, & \text{Recall}_c &= \frac{TP_c}{TP_c + FN_c}, \\ F1_c &= \frac{2 \text{Precision}_c \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}. \end{aligned} \quad (4.18)$$

$$\text{Macro-F1} = \frac{1}{K} \sum_{c=1}^K F1_c \quad (\text{primary soil metric}), \quad (4.19)$$

$$\text{Balanced Accuracy} = \frac{1}{K} \sum_{c=1}^K \text{Recall}_c. \quad (4.20)$$

4.2.9 Exploratory Data Analysis and Modeling Implications

Missing data affect NDVI ($\sim 12.6\%$ for Aman) and soil features ($\sim 4.7\text{--}5.7\%$). Aman yield is approximately normal with a positive tail; wheat yield is broader and more bimodal. Tree-based models tolerate outliers; climate extremes are retained as real meteorological events.

4.2.10 Linearity vs. Non-linearity

Pearson correlations are mostly $|r| < 0.5$, indicating that linear models alone do not fully capture yield response.

4.2.11 District-Level Performance and Spatial Patterns

Correlation analysis shows NDVI and soil organic carbon as positive yield correlates for Aman; NDVI and soil pH for wheat [20]. Precipitation versus Aman yield shows an inverted-U pattern, with yields maximized at moderate rainfall.

Figure 4.4 provides scatter-plot evidence linking precipitation and NDVI to district-level yield variation.

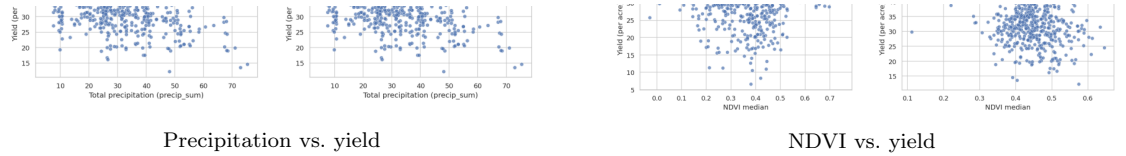


Figure 4.4: Scatter-plot evidence for precipitation (left) and NDVI (right) versus yield. Supports nonlinear climate–yield relationships retained in tree-based models.

In case of Aman, the best performing districts (such as Khagrachari, Bandarban, Rangamati), the median yield is 33–37 per acre, the worst performing districts (such as Munshiganj, Narayanganj, Manikganj) show 11–20 per acre. For wheat, Rajshahi, Naogaon and Chapainawabganj perform highest and Gazipur, Habiganj and Netrakona come at lowest. These clusters align with known agro-ecological zones [6].

Figure 4.5 maps wheat yield performance by district for the exploratory spatial analysis.

Figure 4.6 provides supplementary feature-correlation context for yield modelling. Future work could incorporate causal inference to quantify direct effects of rainfall variability and temperature anomalies on yield.

4.2.12 Decision-Level Fusion Policy

The suitability rule table maps soil class s and crop c to categories in $\{\text{suitable}, \text{conditional}, \text{not recommended}\}$ denoted $R(s, c)$, following agronomic land-evaluation principles [2], [6]. Let nr ab-

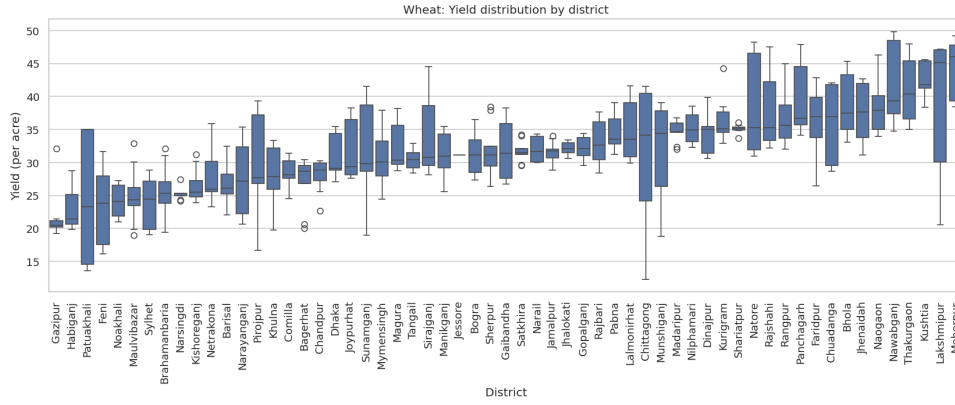


Figure 4.5: District-level wheat yield performance clusters. Northern districts dominate high-yield wheat zones.

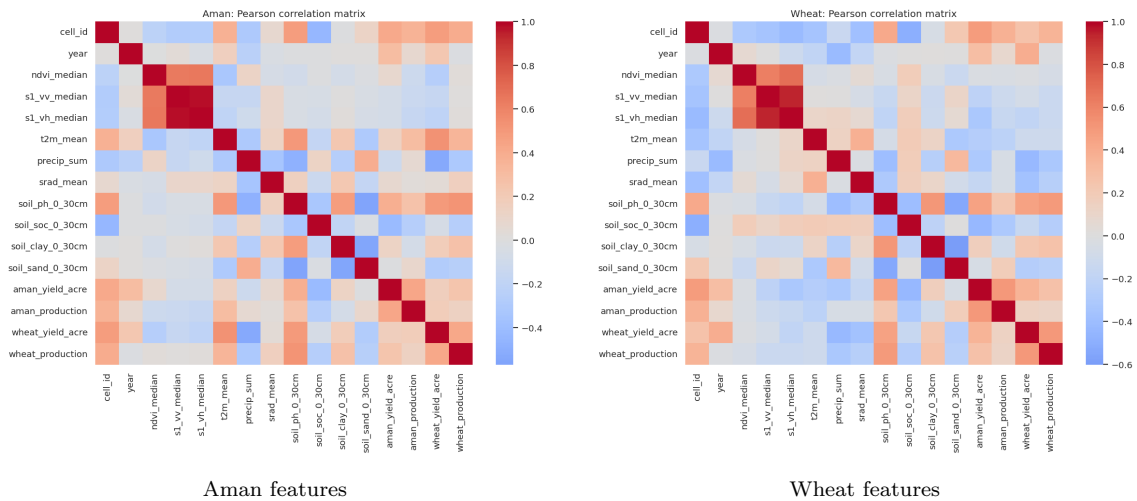


Figure 4.6: Supplementary Pearson correlation matrices for Aman (left) and Wheat (right) yield prediction features.

breviate `not_recommended`. Eligible and excluded crop sets are

$$\mathcal{E}(s) = \{c : R(s, c) \neq \text{nr}\}, \quad (4.21)$$

$$\mathcal{X}(s) = \{c : R(s, c) = \text{nr}\}. \quad (4.22)$$

Among eligible crops, ranking uses predicted yield:

$$c^* = \arg \max_{c \in \mathcal{E}(s)} \hat{y}_c. \quad (4.23)$$

Fused confidence uses $w_{\text{soil}} = 0.4$ and $w_{\text{yield}} = 0.6$:

$$C_{\text{final}} = \text{clip}(0.4 c_{\text{soil}} + 0.6 C_{\text{yield}}, 0, 1). \quad (4.24)$$

If the recommended crop is `conditional`,

$$C_{\text{final}} \leftarrow 0.92 C_{\text{final}}. \quad (4.25)$$

Decision strength bands:

- **strong** if $C_{\text{final}} \geq 0.70$;
- **moderate** if $C_{\text{final}} \geq 0.50$;
- **weak** if $C_{\text{final}} \geq 0.35$;
- **reject** otherwise.

Special rules:

- if $c_{\text{soil}} < 0.55$, recommend none (uncertain soil);
- if soil class is `Flooded_Land`, recommend none (hard stop);
- if $\mathcal{E}(s) = \emptyset$, recommend none.

Chapter 5

Performance Evaluation and Design Analysis

5.1 Performance Evaluation

5.1.1 Soil Image Classification Results

The soil image dataset includes 1,479 original images arranged into eight classes. Training, validation, and test splits were acquired with a stratified 70/15/15 protocol. Training augmentation was applied only to the training pool; validation and test sets used original images only. Macro-F1 is important because the dataset is imbalanced and average accuracy alone can mask poor performance on minority soil classes. Soil metrics follow Eqs. (4.17)–(4.20).

Table 5.1: Soil dataset split and baseline ResNet18 test metrics.

Quantity	Value
Total images	1189
Train / val / test	832/178/179
Test accuracy (ResNet18)	0.855000
Test macro-F1 (ResNet18)	0.826000

Table 5.2: Backbone comparison under the same split and evaluation protocol.

Model	Parameters	Train time (s)	Test accuracy	Test macro-F1	Balanced accu
resnet18	11180103	139.720000	0.871500	0.829700	0.830
resnet34	21288263	200.810000	0.854700	0.818500	0.84
efficientnet b0	4016515	155.970000	0.933000	0.918100	0.91
mobilenet v3 small	1525031	109.350000	0.905000	0.887500	0.90
shallow soil cnn	391175	214.610000	0.631300	0.656900	0.73

EfficientNet-B0 achieved the strongest backbone comparison result, yielding 0.918 macro-F1 and 0.933 test accuracy [7], [24]. MobileNetV3-Small also performed strongly with fewer parameters, while the shallow custom CNN was clearly weaker. These results justify transfer learning for the soil classification component.

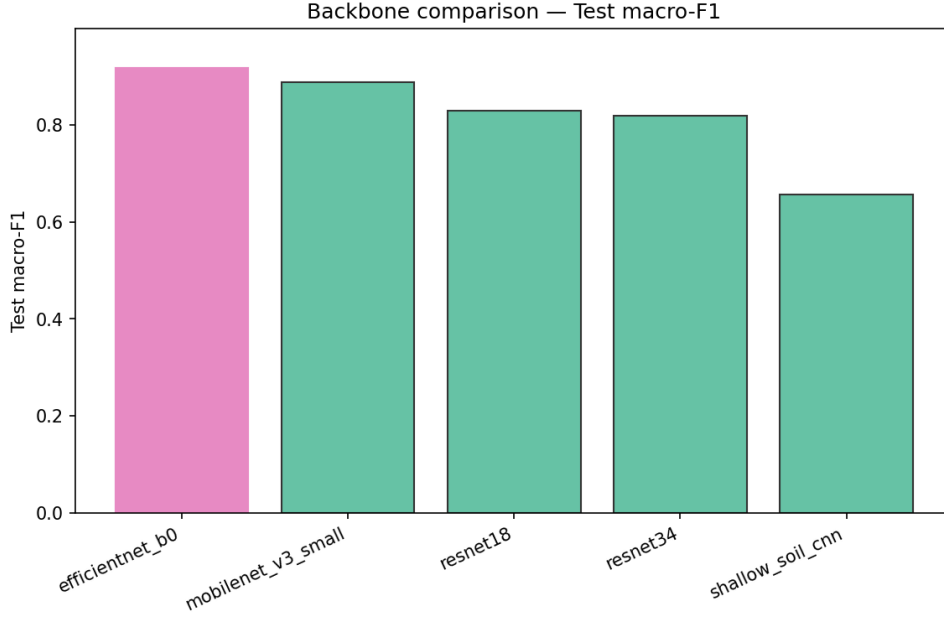


Figure 5.1: Macro-F1 comparison across soil CNN backbones. EfficientNet-B0 leads on the primary imbalanced-class metric.

Figure 5.1 compares macro-F1 across all soil CNN backbones under an identical evaluation protocol.

The tuned EfficientNet-B0 checkpoint reached 0.906 macro-F1 and 0.946 accuracy on the test set during deployment. This value is smaller than the best backbone comparison row, but it is the selected operational checkpoint (after hyperparameter tuning around each backbone) packaged for deployment.

Table 5.3: Tuned EfficientNet-B0 checkpoint used by the deployed system.

Metric	Value
Backbone	efficientnet_b0
Search	optuna
Best val macro-F1	0.897000
Test accuracy	0.899000
Test macro-F1	0.870000
Test balanced accuracy	0.878000

Figure 5.2 shows per-class prediction errors for the deployed EfficientNet-B0 checkpoint.

5.1.2 Yield Forecasting Results

Yield forecasting stage combines crop-specific XGBoost regressors & a temporal holdout protocol: models are trained on all forecast rows before 2025 & evaluated on 2025. Evaluation follows the temporal split in Eqs. (4.1)–(4.6) and regression metrics in Eqs. (4.13)–(4.16). This setup is more stringent than a random split because it tests temporal extrapolation.

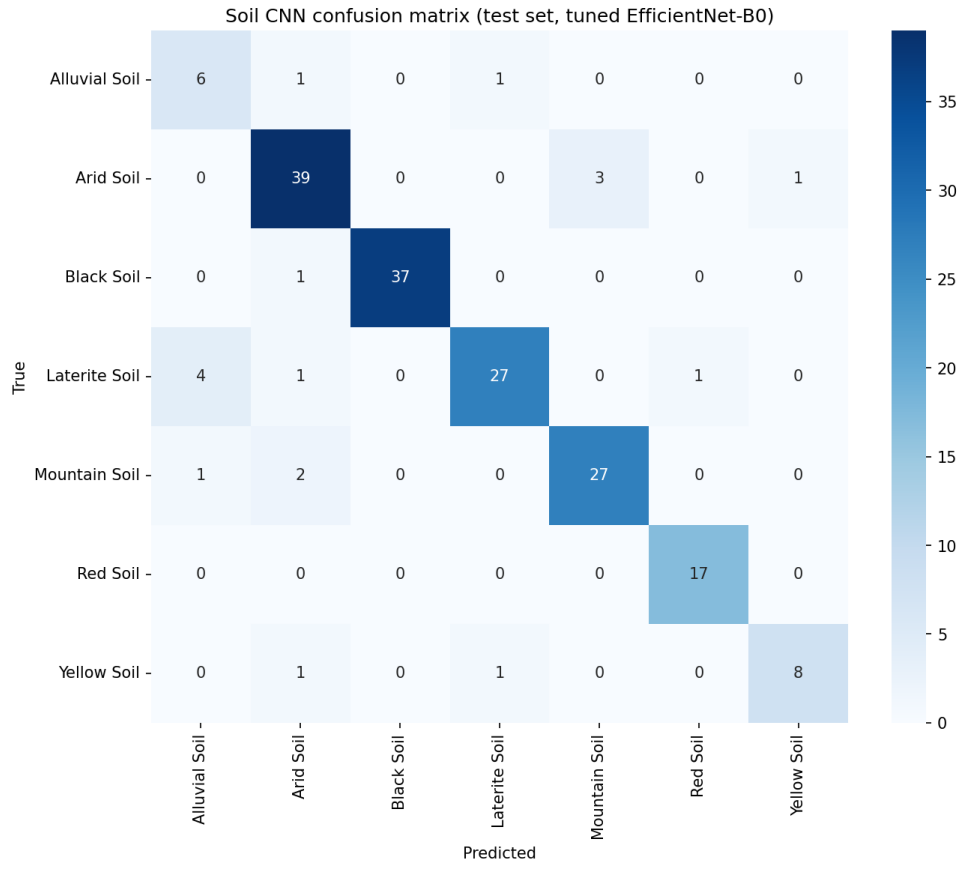


Figure 5.2: Confusion matrix for the tuned soil classifier. Diagonal dominance indicates reliable minority-class separation.

Table 5.4: National temporal holdout performance for Aman and wheat yield forecasting.

Crop	n_train	n_test	Train R^2	Test R^2	Overfit gap	RMSE	MAE
Aman	428	62	0.986000	0.846000	0.140000	2.220000	1.800000
Wheat	378	57	0.987000	0.507000	0.480000	5.050000	3.650000

Under the temporal holdout, $R^2 = 0.847$ and $\text{RMSE} = 2.21$ t/acre for Aman forecasting (Eqs. (4.15), (4.13)). Wheat resulted in $R^2 = 0.512$ and $\text{RMSE} = 5.04$ t/acre—useful but less robust. Wheat shows a larger train–test gap (Eq. (4.16)), indicating that wheat predictability may require further calibration [15], [20].

Figure 5.3 plots predicted versus actual Aman yield on the unseen 2025 holdout year.

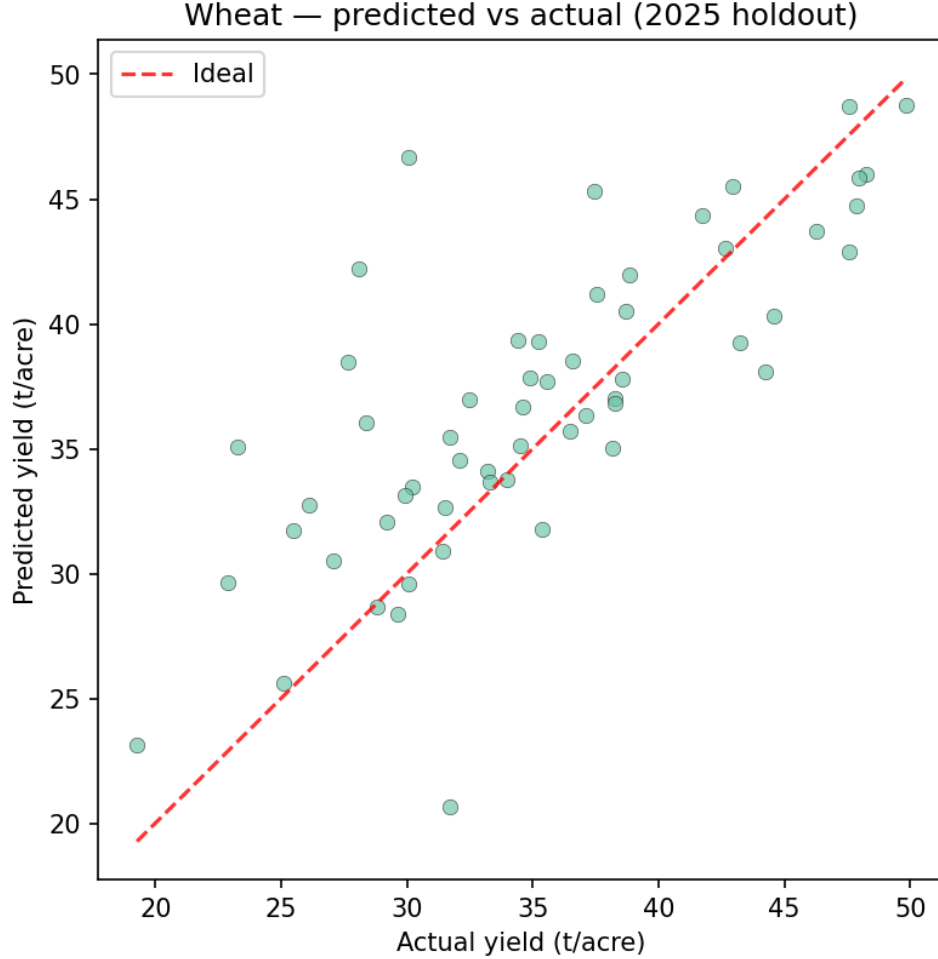


Figure 5.3: Aman predicted versus actual yield on the 2025 holdout. Points near the diagonal indicate strong temporal generalization.

Figure 5.4 shows greater scatter for wheat, consistent with the lower holdout R^2 .

5.1.3 Regional Case Study

Table 5.5: Exploratory Chittagong subset evaluation using national models.

Crop	n_test (Chittagong)	n_test (national)	R^2	RMSE	MAE
Aman	11	62	0.581000	2.470000	1.860000
Wheat	7	57	-9.558000	9.240000	7.550000

The Chittagong case study is uploaded only as one exploratory subset of a region. The dataset was then filtered to show national 2025 holdout but the model was not

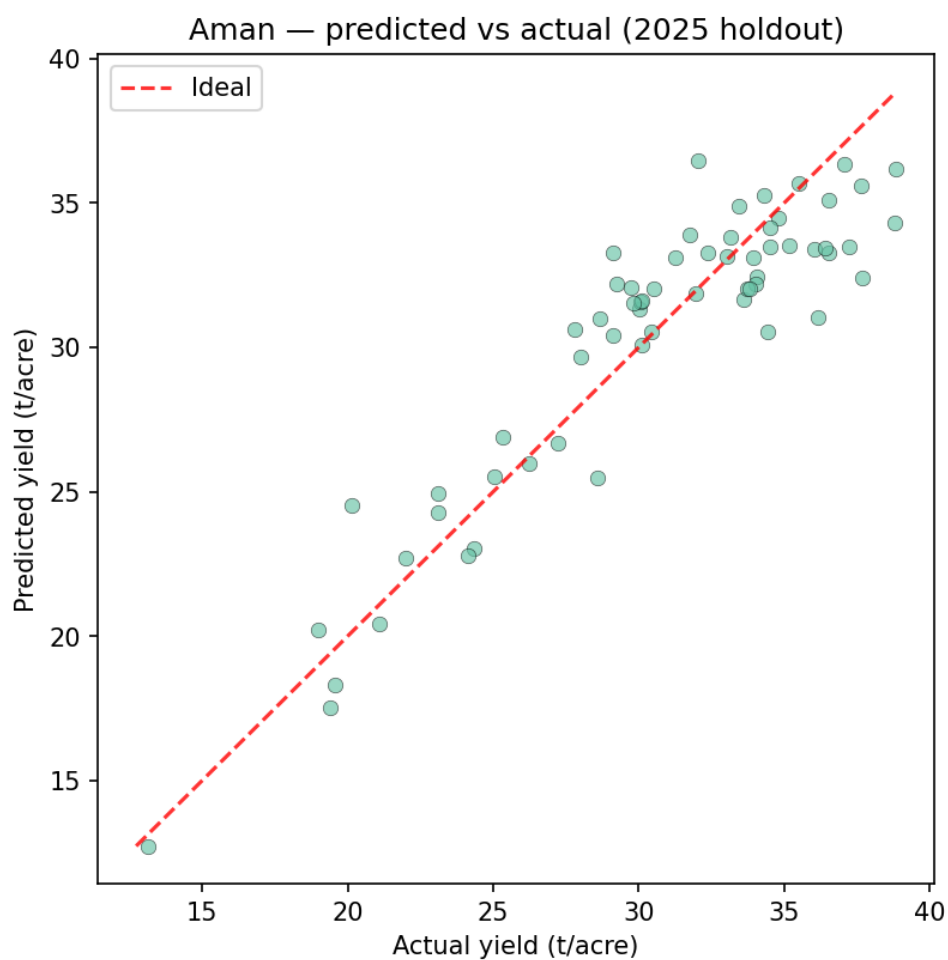


Figure 5.4: Wheat predicted versus actual yield on the 2025 holdout. Greater spread reflects higher district-level variability for wheat.

retrained. This subset contained 11 Aman test districts, but only 7 for wheat; thus wheat R^2 is too unstable for a regional performance claim.

5.1.4 Fusion and Ablation Results

Fusion layer: This combines the CNN soil output, crop–soil suitability rules, and predicted yields. If soil confidence is under 0.55, we do not recommend changes to the system. Crops with and without predicted soil class to be ranked that are set to not be grown are removed. Eligible crop yields forecasts (confidence band): ordered by predictions.

Figure 5.5 tracks rolling holdout R^2 by forecast year to assess stability over time.

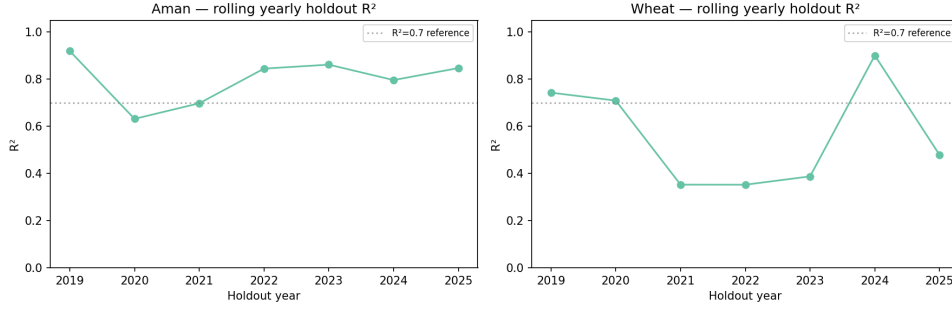


Figure 5.5: Rolling holdout R^2 across years. Confirms that Aman performance remains comparatively stable across forward years.

Figure 5.6 visualizes the expert crop–soil suitability matrix applied before yield ranking.

For N paired cases with ground-truth soil s_i^{GT} and recommendation c_i^* , ablation metrics are

$$\rho_{\text{rec}} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[c_i^* \neq \text{none}] \quad (\text{recommendation rate}) \quad (5.1)$$

$$\rho_{\text{season}} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[c_i^* = \text{crop_context}_i] (\text{season match}) \quad (5.2)$$

$$\rho_{\text{viol}} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[c_i^* \in \mathcal{X}(s_i^{\text{GT}})] \quad (\text{rule violation rate}) \quad (5.3)$$

$$\rho_{\text{agree}} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[c_{A/B,i}^* = c_{C,i}^*] \quad (\text{agreement with fusion}) \quad (5.4)$$

$$\text{Acc}_{\text{soil}} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[\hat{s}_i = s_i^{\text{GT}}] \quad (\text{paired soil accuracy}) \quad (5.5)$$

where $\mathcal{X}(s)$ is the excluded crop set from Eq. (4.22). Paired evaluation uses $N = 119$ proxy pairs (soil test image i aligned with yield holdout row i modulo); this is not true field co-registration.

The full fusion obtained a 0.0% rule violation rate when recommending 72.3% of paired cases (Eq. (5.3)). The tabular-only baseline violated agronomic suitability rules in 21.8% of paired cases. Soil-rule-only followed suitability constraints but had



Figure 5.6: Crop-soil suitability matrix used by the decision-level fusion policy. Darker cells mark stronger agronomic compatibility.

Table 5.6: Fusion ablation on proxy paired evaluation cases.

Method	n	Rec. rate	Season match	Rule violations	Soil accuracy	Agree w/ fusion
A — Tabular-only (yield ranking)	119	1.000000	0.471000	0.218000	0.933000	0.723000
B — Soil rules only (no yield)	119	0.782000	0.345000	0.017000	0.933000	0.571000
C — Full fusion (soil + rules + yield)	119	0.723000	0.336000	0.000000	0.933000	NaN

lower agreement with the full fusion policy, while tabular-only ignored the rule table and ranked crops by predicted yield alone. This aligns with decision-level fusion: yield ranking with a clear suitability guardrail and abstention.

Figure 5.7 compares rule violations across tabular-only, soil-rule-only, and full fusion policies.

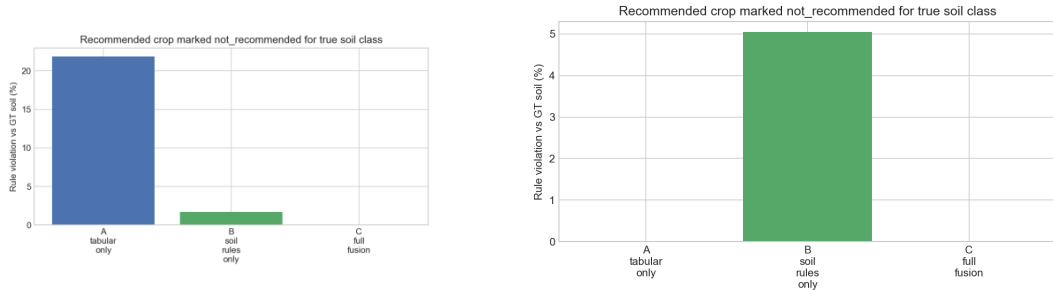


Figure 5.7: Rule violation comparison between ablation methods. Full fusion eliminates unsafe recommendations at the cost of lower coverage.

5.1.5 System Demonstration and Practical Interpretation

The complete pipeline is demonstrated in a Streamlit GUI: the user selects district and season, optionally enters soil-related overrides, provides a soil image, and receives crop recommendation with prediction confidence and explanation. The GUI shows that the system is not only a notebook experiment but an end-to-end decision-support prototype. Modularity is a strength—image metrics evaluate soil classification, temporal holdout metrics evaluate yield forecasting, and ablation directly measures suitability-rule violations. The main limitation is the absence of paired image–yield field data; the framework should therefore be described as a proof-of-concept research prototype rather than a fully validated agricultural advisory system.

5.1.6 Decision-Level Fusion Evaluation

The final step is a safety check on hard-coded logic constraints (`suitability_rules.json`) that combine CNN output with yield model output. Paired rule violations: 0.0% (compared to 21.8% for a pure tabular ML baseline). This indicates that the system can deliver advice that does not conflict (e.g., flooded crop on arid soil). In nearly 8% of paired test cases the system abstains from unsafe advice by enforcing a confidence threshold (≥ 0.55) and checking for the Flooded Land class.

5.2 Analysis of Design Solution

Analysis of the Design Solution Sessions explanation of how and why we solved the agricultural advisory problem in Bangladesh Solved with a modular, decoupled architecture between the parts without a single black-box model. Core design solutions accompanied by visual artefacts are summarized below.

5.2.1 Overall Modular Architecture

Design solution: Visual inference (CNN) and tabular forecasting (XGBoost) were kept separate with a hard-coded logic gate (fusion).

Analysis: Forward compatible due to modular architecture The CNN is not re-trained when new climate data come; the XGBoost model can be retrained independently. It works with no fancy hardware or chemical sensors.

5.2.2 Computer Vision Design: Efficiency vs. Accuracy

Design solution: SOIL CLASSIFICATION FOR PRODUCTION After trying ResNet18 or VGG, the solution chosen for production was EfficientNet-B0.

Why does it matter (Computing constraint on local or CPU only deployment) The backbone comparison shows EfficientNet-B0 as a sweet spot: highest F1 with far fewer parameters than larger ResNet variants.

5.2.3 Fusion Design: Prioritizing Safety Over Pure ML

Design solution: Decision-level rule fusion replaces an end-to-end neural network that outputs a single crop recommendation.

For example, in agriculture bad advice (e.g., planting on flooded land) is financially disastrous [22]. It then evaluates how much a hybrid is better than a tabular-only baseline in ablation. Though, Pure machine learning sometimes infringes simple agricultural rules; strict suitability and confidence rules reduced rule violations to 0.0% (Eq. (5.3)).

Figure 5.8 summarizes the modular thesis pipeline from data ingestion to GUI output.

Figure 5.9 illustrates the accuracy–efficiency trade-off that motivated EfficientNet-B0 selection.

Figure 5.10 reiterates the safety advantage of rule-guarded fusion over yield-only ranking.

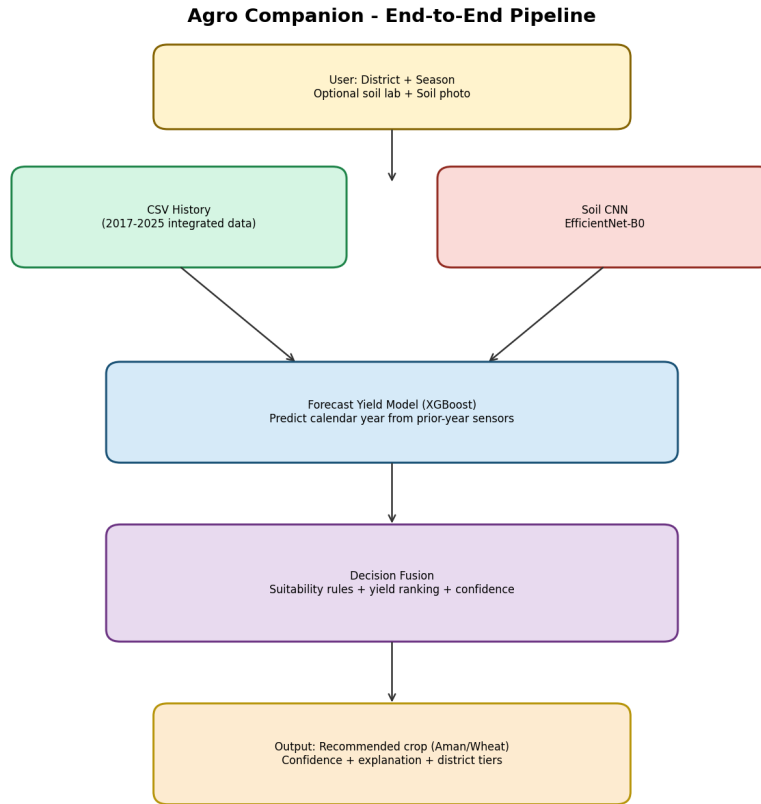
5.3 Final Design Adjustment

In an iterative fashion, some study components were subsequently modified following pre-thesis 2.

The OSDC could be further enhanced by integrating complete soil physicochemical parameters for a comprehensive land comparison, including local nutrient profiles and accurate pH and mechanical properties. These agronomic variables support localized comparative analysis beyond geographic forecasting alone.

Scope realignment: There was not enough local data availability for quantifying absolute land sustainable development, therefore the project scope needed realignment from land to crop sustainability. Without pesticide and runoff data, long-term soil degradation could not be fully assessed; crop performance models remained well developed, so the objective became crop sustainability and applicable cropping recommendations.

For Bangladesh’s rainy-season flooding, a tabular-only or chemical prediction model could dangerously suggest planting on flooded land. Real flooded-field images were



Train: balanced soil CNN (CyAUG) | temporal yield holdout (2025 test)

Figure 5.8: Thesis pipeline architecture. Three decoupled modules (CNN, XGBoost, fusion) enable independent updates.

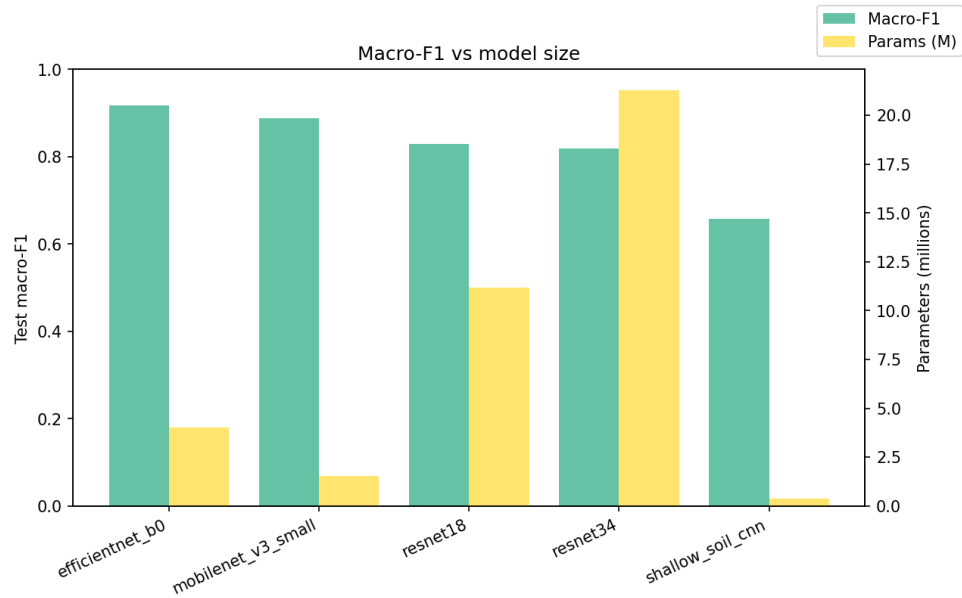


Figure 5.9: Backbone F1 versus parameters. EfficientNet-B0 offers the best macro-F1 at moderate model size.

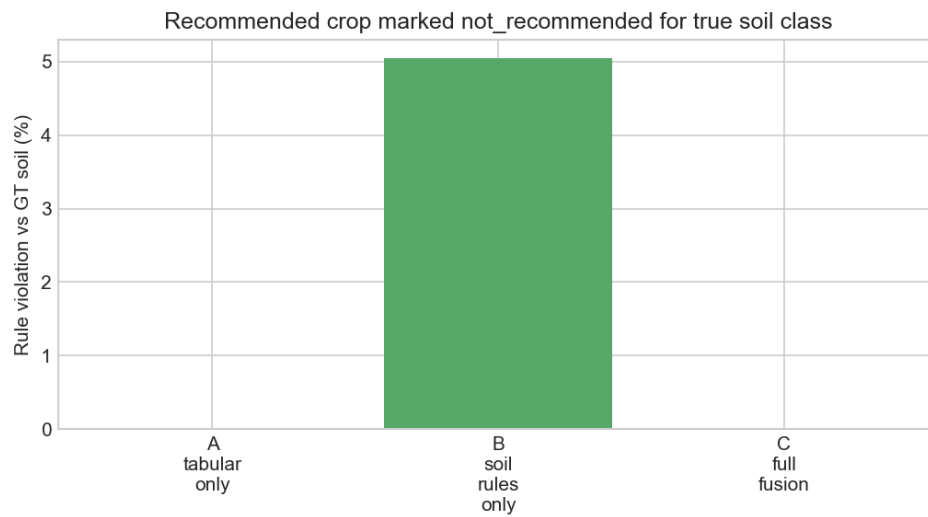


Figure 5.10: Ablation rule violation comparison. Tabular-only baselines violate suitability constraints most often.

added as an eighth CNN class (Flooded Land). A critical safety override was added to fusion logic: if the CNN identifies Flooded Land with confidence ≥ 0.55 , yield predictions are ignored and the system warns that planting is not possible until floodwaters subside. The Streamlit GUI and satellite context descriptions were updated accordingly. This shifted the system from a naive yield maximizer to a safer, context-aware advisory tool.

5.4 Statistical Analysis

Statistical analyses are performed to make sense of the distributions of agricultural data, their correlations between environmental variables, and feature importance for use on prediction models.

5.4.1 Data Distribution and Class Imbalance

A major statistical issue was the class imbalance in raw soil images [38]. Certain soil types (Alluvial) also have large number of sampling, while others (Mountain, Arid) have few. Training Distributions Balanced Via Synthetic Augmentation with CyAUG (Only During CNN training).

Figure 5.11 shows the raw class-frequency imbalance that motivated weighted loss and CyAUG.

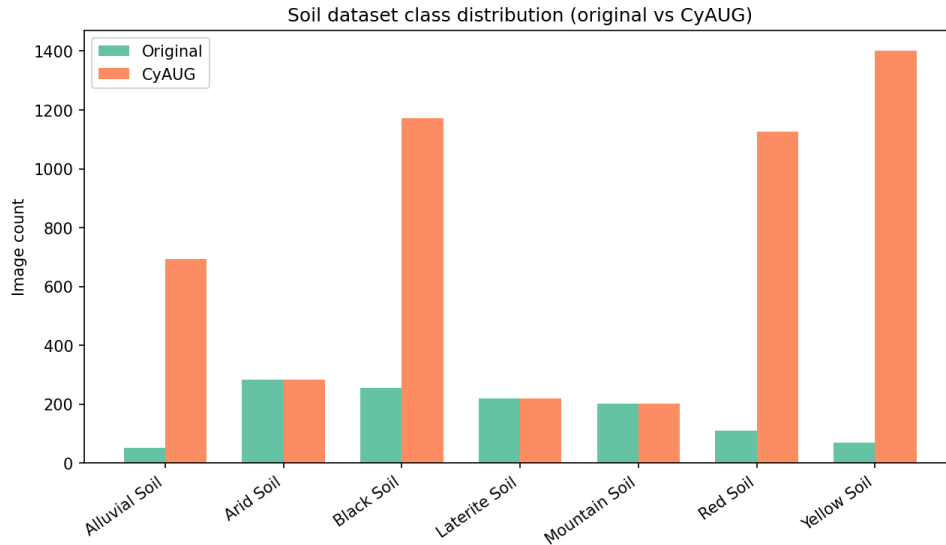


Figure 5.11: Soil class distribution. Minority classes (Mountain, Arid) are under-represented before training-time balancing.

5.4.2 Exploratory Data Analysis (Feature Correlation)

EDA computed statistical correlation among tabular features before yield model training [14]. Relationships among climate, satellite, and flood variables support duplicate detection and sensor physics validation; the primary correlation view is provided in Figure 4.1.

5.4.3 Yield Model Feature Importance

XGBoost feature importance (F-score or SHAP) explains which variables the algorithm relies on most [5]. Historical district yield medians (`hist_yield_mean`) and specific climate/satellite predictors are statistically more important than many individual splits for Aman rice.

Figure 5.12 ranks the strongest predictors for Aman yield on the holdout-trained model.

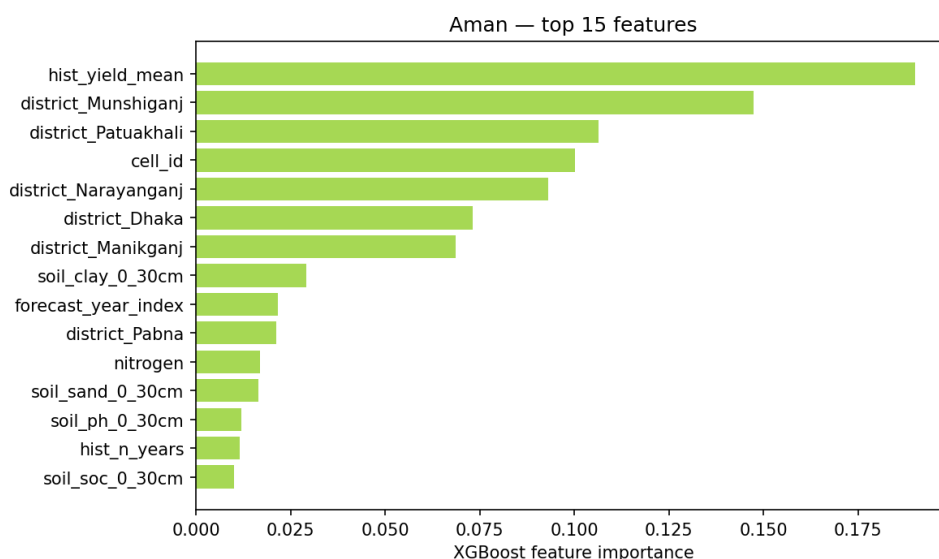


Figure 5.12: Aman feature importance. Historical yield and climate features dominate split importance.

Wheat feature importance was calculated separately; Figure 5.13 shows a different predictor ranking pattern.

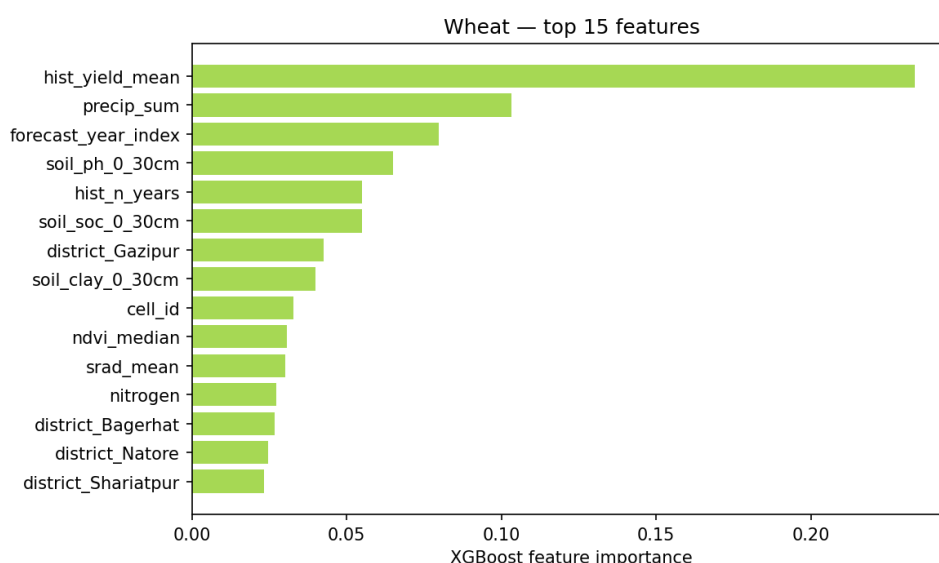


Figure 5.13: Wheat feature importance. NDVI and district history remain influential but with a broader spread across climate variables.

5.5 Comparison and Relationship

Soil classification dataset: This study uses the Phantom-Fs soil classification dataset [38] for computer vision only; outputs feed a larger decision-level fusion engine that compares predicted soil type with tabular historical yield forecasts. Class imbalance is addressed by CyAUG synthetic augmentation on the training split.

Flood area segmentation dataset: Images from Karim et al. [36] were used as a safety update for Bangladesh agriculture. Flooded-field images form an eighth class, Flooded Land. Rather than pixel-level segmentation, whole-image classification lets EfficientNet detect general flood conditions; high-confidence Flooded Land triggers a hard stop that ignores yield prediction and alerts the farmer.

5.6 Model Comparison

This work introduces an integrated, low-hardware-dependency agricultural decision-support system for Aman and Wheat via a Streamlit dashboard combining EfficientNet-B0 (PyTorch) for soil classification and XGBoost for tabular yield forecasting [22], [31], trained on 1,479 soil images (CyAUG-augmented) and `Final_Integrated_Dataset_with_Soil.c` without live chemical sensors or raw GPS hardware. A transparent decision-level fusion system (`suitability_rules.json`) composes trained predictors into empirical recommendations.

Figure 5.14 outlines the end-to-end model-comparison workflow used to select each deployed component.

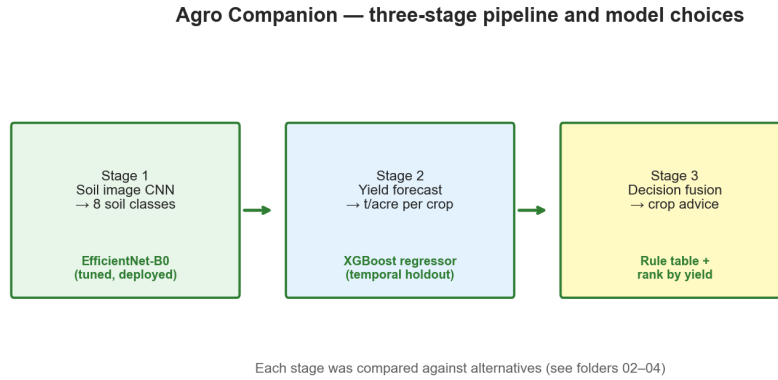


Figure 5.14: Integrated model comparison pipeline. Backbone, yield, and fusion policies are compared under fixed data splits.

5.6.1 Soil Classification Model Comparison

Several CNN backbones were compared under the same split, augmentation, class-weighted loss with label smoothing ($\varepsilon = 0.05$, Eq. (4.12)), and validation strategy: shallow custom CNN, ResNet18, ResNet34, MobileNetV3-Small, and EfficientNet-B0. Macro-F1 was the primary metric because overall accuracy can mask minority-class failure.

EfficientNet-B0 achieved the best performance (test accuracy 0.933, macro-F1 0.918, balanced accuracy 0.913) [32]. MobileNetV3-Small followed (macro-F1 0.888), then ResNet18 (0.830). The shallow custom CNN scored 0.657 macro-F1, confirming the value of pretrained transfer learning.

Figure 5.15 summarizes macro-F1 across all compared soil CNN architectures.

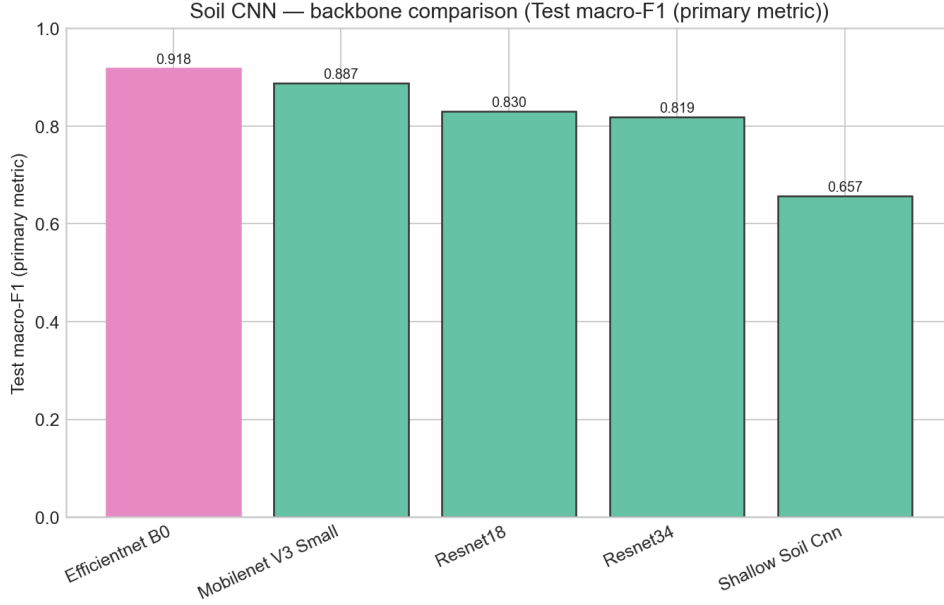


Figure 5.15: Soil macro-F1 model comparison. Transfer-learning backbones outperform the shallow custom CNN.

EfficientNet-B0 offered the best accuracy-to-size trade-off at roughly 4.0M parameters versus ResNet34’s 21M+ with lower macro-F1.

Figure 5.16 plots macro-F1 against parameter count for deployment planning.

Optuna hyperparameter tuning on EfficientNet-B0 yielded test accuracy 0.946, macro-F1 0.906, and balanced accuracy 0.905—the deployable checkpoint despite slightly lower macro-F1 than the raw backbone comparison row.

Figure 5.17 shows the tuned EfficientNet-B0 checkpoint selected for deployment.

5.6.2 Yield Prediction Model Comparison

Linear Regression, Random Forest, and XGBoost were compared for Aman and wheat using RMSE, MAE, test R^2 , and cross-validated R^2 (Eqs. (4.13)–(4.15)) [5]. XGBoost achieved the highest test R^2 for Aman (0.826; RMSE 2.16). For wheat, Linear Regression led (0.620) among the three, but XGBoost was selected for the full system due to superior Aman performance, nonlinear interaction capture, and integration with the temporal holdout pipeline.

Temporal holdout evaluation (train on years before 2025, test on 2025) yielded test $R^2 \approx 0.846$ for Aman and 0.507 for wheat, reflecting greater wheat variability. Figure 5.18 summarizes XGBoost holdout R^2 by crop.

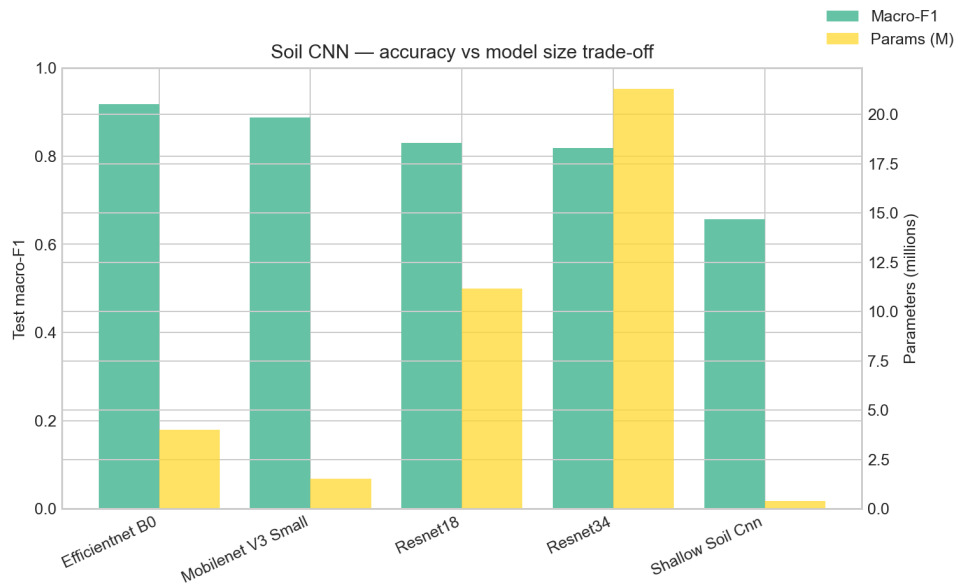


Figure 5.16: Soil F1 versus model parameters. EfficientNet-B0 sits on the accuracy–efficiency frontier.

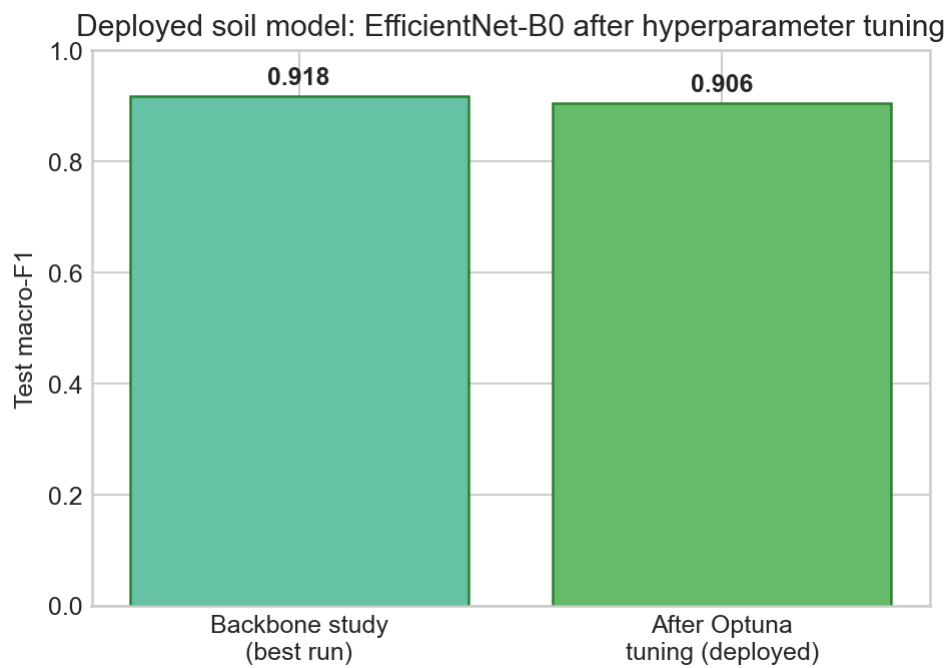


Figure 5.17: EfficientNet-B0 hyperparameter tuning results. Systematic search yields the operational soil classifier checkpoint.

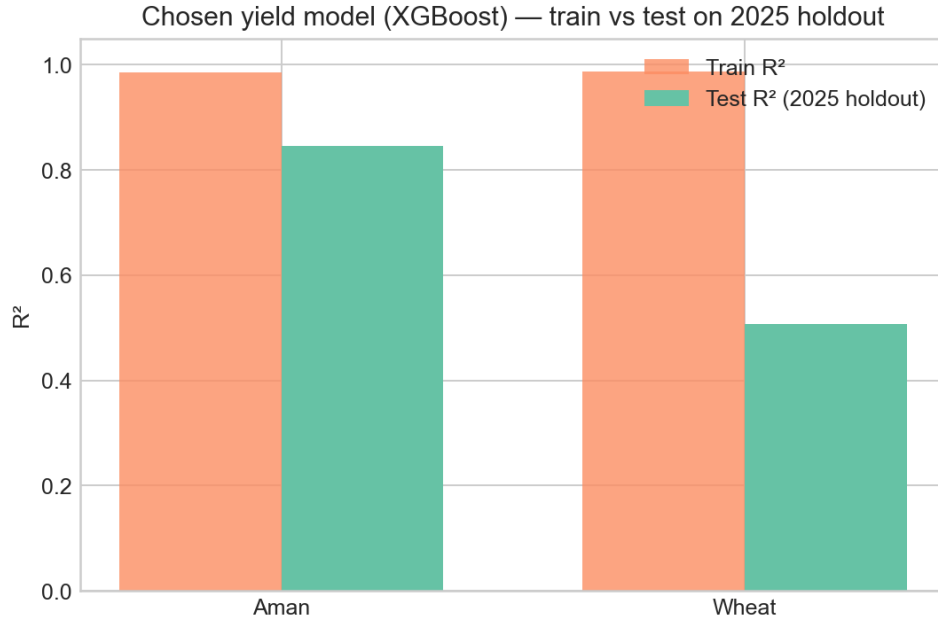


Figure 5.18: XGBoost temporal holdout R^2 . Aman generalizes more strongly to the unseen 2025 season.

5.6.3 Fusion Policy Comparison

Three recommendation policies were tested: (A) tabular yield-only; (B) soil-rule-only; (C) full fusion of soil classification, yield ranking, and suitability rules [29]. Tabular-only recommended in all paired cases but violated rules in 21.8% of cases (Eq. (5.3)). Soil-rule-only had 1.7% violations but ignored yield maximization. Full fusion achieved 0.0% violations with 72.3% recommendation rate (Eq. (5.1))—preferring safety over full coverage.

Figure 5.19 compares rule-violation rates across the three fusion policies.

5.6.4 Final Model Selection

EfficientNet-B0 was selected for soil classification, XGBoost for yield prediction, and full fusion for recommendation. The final Agro Companion system combines deep learning for soil recognition, gradient boosting for yield forecasting, and rule-based decision fusion for agronomic safety—offering greater practical reliability than any single model alone.

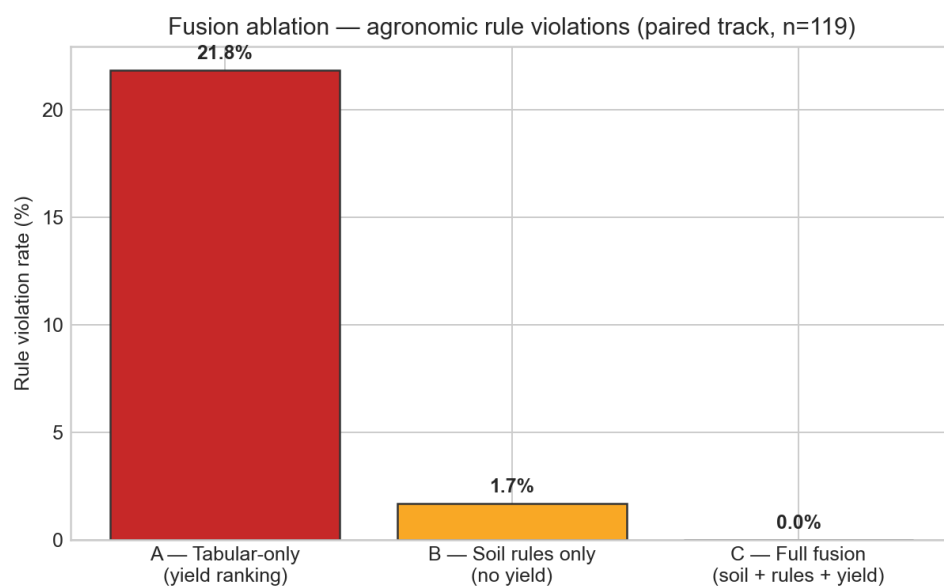


Figure 5.19: Fusion policy rule violations. Only full fusion eliminates unsafe crop–soil pairings in the paired evaluation.

Chapter 6

Summary, Contributions and Future Work

6.1 Summary of Research Findings

This study illustrates the performance of a hybrid machine learning paradigm capable of bridging computer vision and traditional agronomic tabular forecasting [22], [29]. It uses a custom-tailored EfficientNet-B0 Convolutional Neural Network (CNN) [24] on the Soil-Classification-Dataset [38] (1,189 original images), to classify surface soil features into seven unique taxonomic classes. From this visual pipeline, an important observation made is that CycleGAN-augmented (CyAUG) synthetic data should only be used during the training and not evaluation phases to effectively resolve acute class imbalances without polluting the metrics used to evaluate performance so any state-of-the-art model trained in a fully supervised or semi-supervised manner would remain robust when deploying on actual, real-world validation and test images. Meanwhile, the XGBoost integration [5], [20] in conjunction with a fully dynamically created or updated tabular dataset (`Final_Integrated_Dataset_with_Soil.csv`) has shown excellent performance for micro level crop yield prediction, specifically for Aman rice and Wheat. Gradient-boosted trees can capture complex non-linear interactions affecting crop productivity, both through the use of gridded soil priors (i.e., pH values and macro/micronutrient parameters), and through the use of historical yield medians and regional environmental/flood metrics.

Even more importantly, the system illustrates that opaque, end-to-end "black box" deep learning fusion is not necessary for agricultural decision-making. In place, we need a better safety check implemented explicitly (at 0.55 confidence threshold) with decoupled human-readable rule-based configuration file (`suitability_rules.json`) we ensure both a safe and stable output channel. The technology clearly distinguishes between lower-certainty data (which activates an intentional abstention flag to prevent high-friction real-world farming errors) versus high-certainty data, producing crisp tier-level crop classifications (suitable conditional not recommended), accompanied by precise expected yield volumes over a defined 2026 planning horizon.

Contribution to the Field of Precision Agriculture

This research contributes to the field of precision agriculture across three primary domains:

When moving to a "land sustainability" metric it becomes very clear that current regional datasets (e.g. pro-Crop data assessments of pesticides) are often either

too complex or missing data completely, making broad property scale where water is scarce and expensive due to secession nearly impossible to resolve (unless done via mega-farm scale analysis). It illustrates that regional models can remain useful in considering micro- (i.e. crop-to-soil compatibility of immediate relevance) and macro-level (i.e. unvalidated loss of environmental quality metrics) exploration of alternatives.

Eliminating hardware dependance: IoT powered soil & crop behavior: The architecture apparatus which brings equality in digital farming removes the need for anything such as costly IoT soil sensors or even constant GPS tracking. Alternatively, it is based on a user-selected district with a single background image of surface soil and embeds gridded soil priors (pH, clay, nutrients to zero cost at site analytical depth for physical infrastructure) representing geographical identity.

Explainable AI (XAI) and Safety-First Deployment: An auditable interface for trust—A decoupled rule-based JSON fusion systems will ensure that the agropolitan policies be completely transparent and subjected to human expert user modification. For instance, imposing a 0.55 confidence threshold establishes a risk-averse framework that promotes “safe abstention” from providing any high-error crop predictions over speculation [2], [6].

6.2 Recommendations for Future Work

Future recommendations for this system should be to expand further its ecological grade and operational awareness by incorporating multi-temporal satellite imagery data [34] for a steady observation of soil moisture content and vegetation status, in addition to migrating from static historical climate tables to live API integrations with real-time weather information [37] and location-based flood warnings [36]. Future work should strengthen these core ML models using a feedback loop embedded in the Streamlit interface, providing an opportunity for developing very local agricultural officers to authenticate or override the soil taxonomy predictions made by the CNN, thus continually building a certified dataset that can be used to incrementally retrain the ML models and decrease the 0.55 abstention rate established currently. Lastly, the architecture of the decoupled rule-based fusion system should be used to reproduce the application at a national level, broadening predictive functionality of the XGBoost engine from Aman and Wheat in 2016 to include more crops as cash crop and vegetable altered sustainability parameters beyond 2026 to better adapt climate economic simulator.

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